Supplementary Information

Assessment of computational methods for diabetic retinopathy lesion segmentation in fundus images

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Supplementary Notes

Supplementary Note 1: the registered model variants

One entry per registered variant: family, tier (1, main comparisons; 2, as time allows; 3, paper audit only), reproducibility level (R0 no code; R1 model definition only; R2 inference code and weights; R3 training code; R4 training code, weights, splits and environment), track (A, released weights; B, unified retraining of the official code; C, re-implementation from the paper), the code used, and what was changed or found when the code was run. The registry is `bench/configs/models.yaml` in the repository; the notes are those recorded at the code audit and updated as runs completed.

General segmentation networks..

- **U-Net** (tier 1, reproducibility R3, track B); no author repo; use a reference PyTorch implementation.
- **DeepLabv3** (tier 1, reproducibility R3, track B); code: `official_code/DeepLabv3plus`; registered name is DeepLabv3. torchvision deeplabv3_resnet101 is ASPP plus bilinear upsampling, not the DeepLabv3+ decoder. The TensorFlow repo is the paper code and is not the B1 network..
- **HRNet** (tier 1, reproducibility R3, track B); code: `official_code/HRNet-Semantic-Segmentation`.
- **nnU-Net** (tier 1, reproducibility R4, track B); code: `official_code/nnUNet`; shallow clone 2026-10-04; train entry nnunetv2/run/run_training.py.
- **Swin-Unet** (tier 1, reproducibility R3, track B); code: `official_code/Swin-Unet`.
- **FCT** (tier 1, reproducibility R3, track B); code: `official_code/FCT`; official PyTorch port in PyTorch/main.py, not the TensorFlow entry. B1 adaptation widens the stem to 3 channels, assigns the four head channels to MA HE EX SE, and sets forward_size 384..
- **H2Former** (tier 1, reproducibility R3, track B); code: `official_code/H2Former`; general medical segmentation model used on several organs. Not a DR-specific method. B1 uses idrid_train.py for both IDRiD and DDR..
- **UNet++** (tier 2, reproducibility R3, track B); code: `official_code/Nested-UNet`.
- **UPerNet** (tier 2, reproducibility R3, track B); code: `official_code/UPerNet`.

Fundus-specific: multi-scale CNNs..

- **M2MRF** (tier 1, reproducibility R4, track AB); code: `official_code/M2MRF`; mmseg 0.8 / torch 1.6; weights on Google Drive; README: paper reports mean of 3 runs, code reports best.
- **HACDR-Net** (tier 1, reproducibility R3, track B); code: `official_code/HACDR-Net`; mmseg 0.30 / mmcv-full 1.6.2; no released weights found.
- **HRDecoder** (tier 1, reproducibility R4, track AB); code: `official_code/HRDecoder`; added 2026-10-14 (docs/paper/18): MICCAI 2024, mmseg 0.16 / mmcv 1.7, HRNet-W48 backbone; weights on Google Drive (IDRiD 1tRBNqpbFwnNBGQBMw8UYL8sjJz DDR 1CBTHOtxToHQ13H4SZIUBZTwX0zwmIduh); runs in the retiseg-hacdr environment.
- **WSRFNet** (tier 1, reproducibility R1, track AB); code: `official_code/WSRFNet`; checked 2026-10-03: model.py + test.py, no train.py; weights by email to xuanli@stu.hit.edu.cn. 2026-10-14: tier 1, the training loop and the paper's loss are in bench/models/wsrfnet.py (docs/paper/18).
- **L-Seg** (tier 2, reproducibility R1, track C); code: `official_code/L-Seg`; Caffe prototxt and loss layer.
- **SAA** (tier 2, reproducibility R0, track C); code: `official_code/SAA`; official repo empty.
- **CARNet** (tier 3, reproducibility R0).

- **PMCNet** (tier 1, reproducibility R3, track B); code: `official_code/PMCNet`; train.py present; TF 1.13/Keras 2.3; no released weights; bundled data is one sample image. 2026-10-14: tier 1, rewritten layer by layer in PyTorch (bench/models/pmcnet.py, docs/paper/18); the released dataGenerator one-hot-encodes 3 classes against the 5-class head.
- **MLNet** (tier 3, reproducibility R0).
- **LezioSeg** (tier 3, reproducibility R0).
- **HEDNet-cGAN** (tier 3, reproducibility R0).
- **Guo2024** (tier 3, reproducibility R0); Google Drive link of unknown content.
- **CLC-Net** (tier 1, reproducibility R0, track C); code: `reproductions/Wang2023_CLC-Net`; no author code; re-implemented from the paper by the user (assumptions A1-A22 in the reproduction), B-std through bench/models/clcnet.py (docs/paper/18).

Fundus-specific: relational and transformer models..

- **RTNet** (tier 2, reproducibility R0, track C); image + vessel input.
- **VTA** (tier 3, reproducibility R0).
- **TC-Net** (tier 1, reproducibility R0, track C); code: `reproductions/Zhang2023_TC-Net`; no author code; re-implemented from the paper by the user (assumptions in the reproduction), B-std through bench/models/tcnet.py (docs/paper/18).

Fundus-specific: semi-supervised..

- **SSMD-UNet** (tier 3, reproducibility R0); author split 70/5/25; EyePACS pretraining overlaps Retinal-Lesions source.
- **Zhou2019-CL** (tier 3, reproducibility R0).

Foundation models..

- **SAM** (tier 1, reproducibility R2, track B); code: `official_code/segment-anything`.
- **SAM2** (tier 1, reproducibility R3, track B); code: `official_code/sam2`.
- **SAM3** (tier 1, reproducibility R3, track B); code: `official_code/sam3`; shallow clone 2026-10-04; train entry sam3/train/train.py; checkpoints are gated on Hugging Face.
- **MedSAM** (tier 1, reproducibility R3, track B); code: `official_code/MedSAM`.
- **SAMed** (tier 1, reproducibility R3, track B); code: `official_code/SAMed`.
- **SAM-Adapter** (tier 1, reproducibility R3, track B); code: `official_code/SAM-Adapter`.
- **SAM2-UNet** (tier 1, reproducibility R3, track B); code: `official_code/SAM2-UNet`.
- **RETFound** (tier 1, reproducibility R3, track B); code: `official_code/RETFound`; rmaphoh/RETFound_MAE, shallow clone 2026-10-04; main_finetune.py is classification. UPerNet head is still to be added. Weights on Hugging Face.
- **PerSAM** (tier 2, reproducibility R2, track B); code: `official_code/PerSAM`.
- **TP-DRSeg** (tier 3, reproducibility R0); code: `official_code/TP-DRSeg`; paper link 404; mirror has README only.
- **SAM2-DR-Chen2025** (tier 3, reproducibility R0).
- **Ophthalmology-SAM** (tier 3, reproducibility R0).

Supplementary Note 2: the training budget, the selection rule and the disclosed revisions

Budget. The training amount of the unified protocol is counted in canvas pixels: one equivalent epoch draws as many patch pixels as the canvases of the training split contain. The number of equivalent epochs was frozen from the validation curves of two models (U-Net and M2MRF) on each source before the first main-table run: 1,000 on IDRiD (44 training images) and 400 on DDR (383). The rule for the freeze was written down first: the smoothed validation curve of both probe models had to reach its maximum before 90% of the schedule, otherwise the amount would be doubled once and the probe repeated. Both probes met the rule at these values; a later two-fold probe on three cells (HRNet-W48 on DDR, HRDecoder on both sources; main text, case study) is reported as a sensitivity analysis and did not change the budget.

Selection. A run is validated 20 times at equal intervals on the validation split of its training source and the checkpoint with the highest validation mAUPR inside the field of view is scored; the last step is kept for sensitivity. The test split is never read during training. nnU-Net keeps its own schedule and selection.

Revisions after a test number had been read. One: the reference U-Net's source recipe, a binary cross-entropy plus a sigmoid Dice term computed per image with smoothing 1, drove the rare-class channels to zero on DDR. The revision (Dice pooled over the batch per class, smoothing 10^{-5}) was decided from the training and validation curves of a diagnostic run, after the registered recipe's test numbers had been read; both versions are reported (Supplementary Table 6). Before any foundation model was trained the same rule was registered for SAM2-UNet and MedSAM, whose per-image terms share the mechanism, and it fired for both on DDR; their registered runs are reported next to the revised ones. No budget, augmentation or selection rule was changed after a test number had been read.

Scope changes. The source-recipe track (each model trained with its source recipe on the shared data) was stopped for compute on 2026-10-11 before any of its six started runs finished, and no number from it is reported. The author-recipe reruns were reduced to M2MRF (IDRiD) and HRDecoder (IDRiD); two further M2MRF seeds were added on 2026-10-25 when cards were free. The learning-rate sweep ran on DDR for HRNet-W48 and M2MRF only. The encoder comparison's IDRiD column (three cells) was added after the DDR result of the first pair had been read. The reference model's two cells and their ensemble were registered on 2026-10-23, before any merged run.

Supplementary Note 3: evaluation protocol

Pixel level. For each class, the positive and negative pixels of all test images inside the field of view (and outside the ignore region of that class) are pooled, and AUPR is computed exactly over the 16-bit probability levels of the stored maps; this equals the average precision over the pooled pixels. The 11-threshold AUPR of the evaluation code released with M2MRF is computed on the same maps and reported next to it. Dice and intersection over union are given at the threshold that maximises the mean Dice on the validation split of the training source (val-opt) and at 0.5.

Lesion level. Lesions are 8-connected components. A ground-truth lesion is detected if any predicted positive pixel overlaps it; a predicted component is a true positive if it overlaps any ground-truth pixel of the class. Lesion size is the equivalent diameter of the component relative to that of the field of view; the small/medium/large boundaries, 0.005 and 0.011, are the medians of microaneurysms and hemorrhages on the training images of IDRiD and DDR, frozen before training.

Intervals and tests. 95% intervals are percentile intervals of 2,000 bootstrap replicates over test images; replicates pool per-image histograms coarsened to 4,096 bins. Two models are compared by a paired bootstrap with shared image indices and by a two-sided Wilcoxon signed-rank test on per-image Dice with Holm correction. Rankings across the eight test-set blocks use the Friedman test and the Nemenyi critical difference; the DiaRetDB1 block uses the official image-level AUC on DiaRetDB1.

Ultra-widefield images. The TJDR ultra-widefield set is predicted at native resolution with sliding windows for every model trained on a source with a known field of view; for cells whose training sources fix no single field of view (the factor cells trained on TJDR standard-field images, on FGADR or on merged sources), only the version resized to the 1,440-pixel canvas is run and stands for the result on that set.

Supplementary Tables

Supplementary Table 1 Literature audit: the experimental setup stated by each method paper in the ledger. Split, input resolution, AUPR implementation and the provenance of baseline numbers as extracted from the paper's text and tables (literature/meta/paper_audit.tsv in the repository holds the verbatim evidence for every cell).

Paper	Year	Venue	Datasets	IDRiD split	DDR split	Input resolution	AUPR impl.	Pooling	Baselines	*n*
Guo2019 L-Seg	2019	Neurocomputing	IDRiD;e-ophtha;DDR	official	official	IDRiD and e-ophtha: 'resize each image and…	11-threshold	not stated	rerun all	4
Zhou2019 collaborative semi s…	2019	CVPR	IDRiD	official	NA	640x640 (Fig. 2 shows 640*640*3 input / 640…	not stated	not stated	not stated	5
Xiao2020 HEDNet cGAN	2020	arXiv (2007.138…	IDRiD	official	NA	original 4288 x 2848; training: 'we randoml…	exact	pooled ('…	rerun all	2
Bo2022 SAA	2022	ISBI	IDRiD;DDR	official	official	IDRiD: 'resized images and their correspond…	not stated	not stated	not stated	6
Guo2022 CARNet	2022	Complex & Intel…	IDRiD;e-ophtha;DDR	official	official	'all images are uniformly resized to 1024 x…	NA	not stated	rerun all	8
Hasan2022 DRG-Net	2022	arXiv (2212.146…	IDRiD;FGADR	official	NA	'For all experiments, the sizes of the inpu…	not stated	not stated	copied	8
He2022 PMCNet	2022	arXiv (2205.157…	DDR;IDRiD;e-ophtha	official	official	IDRiD and E-ophtha: 'resized each image and…	not stated	not stated	rerun some	11
Huang2022 RTNet	2022	IEEE Transactio…	IDRiD;DDR	official	official	training: large images random resized and c…	exact	not stated	rerun some	7
Ali2023 LezioSeg	2023	Electronics (MD…	IDRiD;e-ophtha;DDR	official	official	training: each image split into 4 non-overl…	per-image mean	per-image	not stated	11
Geetha2023 RILBP-YNet	2023	Biomedical Sign…	IDRiD;e-ophtha (plu…	official	NA	black border cropped by contour analysis, t…	11-threshold	not stated	copied	11
Liu2023 M2MRF	2023	Pattern Recogni…	DDR;IDRiD	official	official	DDR: downsample so long side = 1024, zero-p…	11-threshold	pooled	rerun some	12
Qiu2023 Learnable Ophthal-molo…	2023	arXiv	IDRiD	not stated	NA	not stated	NA	not stated	not stated	1
Ullah2023 SSMD-UNet	2023	Sci Rep	FGADR;IDRiD	official	NA	'bicubic interpolation to resize the images…	not stated	not stated	rerun some	9
Wang2023 CLC-Net	2023	Neurocomputing	IDRiD;e-ophtha;DDR	official	official	no global resize: Otsu to remove non-tissue…	not stated	not stated	rerun some	8
Wang2023 VTA hyperbolic	2023	Scientific Repo…	IDRiD;DDR	official	official	not stated (ViT patch size 16; model pre-tr…	not stated	not stated	rerun all	4
Yin2023 DB-UNet	2023	IEEE Access	IDRiD;DDR;e-ophtha…	official	official	IDRiD: ROI 2560x3840 downscaled by t (t=2 f…	11-threshold	not stated	rerun some	8
Zhang2023 TC-Net	2023	Comput Biol Med	IDRiD;DDR;ISIC-2017	not stated	own	all images resized to 512x512	NA	not stated	not stated	7
Bian2024 MLNet	2024	Algorithms (MDP…	DDR;DiaRetDB1;IDRiD	official	official	black borders cropped, images uniformly res…	not stated	not stated	rerun some	7
Fu2024 LFRC-Net	2024	Applied Science…	IDRiD;DDR;Local ult…	official	official	IDRiD: cropping, zero-padding and resizing…	not stated	not stated	rerun some	12
Guo2024 deep multiscale	2024	SSRN preprint (…	IDRiD;DDR;DiaRetDB1…	official	official	three input scales 384x384, 512x512, 1024x1…	not stated	pooled	not stated	5
Li2024 TP-DRSeg	2024	MICCAI	IDRiD;DDR	not stated	not stated	resized to 1024x1024 ('We resize the input…	not stated	not stated	not stated	8
Li2024 WSRFNet	2024	IJCAI	IDRiD;DDR	official	official	resized to 1440x960 (IDRiD) and 1024x1024 (…	11-threshold	pooled	rerun some	8
Playout2024 cross dataset gen…	2024	arXiv (ISBI-sty…	IDRiD;DDR;FGADR;Ret…	official	official	cropped to remove black background, resized…	11-threshold	not stated	copied	3
Xu2024 HACDR-Net	2024	AAAI	IDRiD;DDR	official	official	IDRiD resized to 1440x960 (following M2MRF)…	not stated	not stated	not stated	13
Xu2024 MSAG	2024	Scientific Repo…	IDRiD;DDR	official	not stated	IDRiD: 4288x2848 cut into 54 blocks of 512x…	not stated	not stated	not stated	9
Xu2024 PGAFT	2024	Sci Rep	IDRiD;DDR	official	official	'both the larger color fundus images and va…	not stated	not stated	rerun all	11
Chen2025 SAM2 DR	2025	ISBI	IDRiD;DDR	official	official	resized to 1024x1024 ('Input images are res…	not stated	not stated	not stated	6
Guo2026 TVFNet	2026	Neural Comput A…	IDRiD;DDR;FGADR;e-o…	official	official	'the fundus images were uniformly resized t…	not stated	not stated	rerun some	18
Qi2026 TEDR-YNet	2026	IEEE Access	IDRiD;DDR;TJDR	official	own	images from all datasets were resized to 51…	not stated	not stated	rerun some	10
Wang2026 CSIF	2026	Applied Soft Co…	DDR;IDRiD	official	official	DDR: long edge scaled to 1024 then zero-pad…	not stated	not stated	rerun some	11

"not stated" means the paper gives no information. Baselines: how the comparison numbers were obtained (rerun, copied, mixed, not stated). *n*, number of compared methods.

Supplementary Table 2 One method, many reported values. Every method that appears with a mean AUPR on IDRiD or DDR in at least two papers of the ledger: the number of papers, the lowest and highest value, their spread, and how many of the values were produced by the reporting paper (rerun), carried from an earlier paper (copied) or cite a source outside the ledger (unverifiable).

Method	Dataset	Papers	Min	Max	Spread	Rerun	Copied	Unverif.
AIMIA	IDRiD	3	.437	.437	.000	0	0	0
Attention U-Net	DDR	2	.198	.389	.191	2	0	0
Attention U-Net	IDRiD	2	.418	.549	.132	2	0	0
CASENet	DDR	3	.193	.193	.000	1	0	0
CASENet	IDRiD	4	.482	.482	.000	1	0	0
CE-Net	DDR	2	.309	.504	.195	2	0	0
CE-Net	IDRiD	2	.470	.635	.165	2	0	0
CIL	IDRiD	3	.535	.535	.000	0	0	0
DNL	DDR	2	.401	.401	.000	1	0	0
DNL	IDRiD	2	.591	.591	.000	1	0	0
Deep-multiscale (Guo2024)	DDR	3	.349	.353	.004	0	0	0
Deep-multiscale (Guo2024)	IDRiD	2	.710	.710	.000	0	0	0
DeepLabv3+	DDR	10	.290	.452	.162	6	0	0
DeepLabv3+	IDRiD	11	.483	.666	.183	6	0	0
DenseUNet	DDR	2	.336	.483	.147	1	0	0
DenseUNet	IDRiD	2	.564	.651	.087	1	0	0
EAD-Net	IDRiD	2	.549	.549	.000	0	0	0
FCRN	DDR	2	.096	.096	.000	1	0	0
FCRN	IDRiD	3	.455	.455	.000	1	0	0
FCT	DDR	2	.250	.250	.000	0	0	0
FCT	IDRiD	2	.431	.431	.000	0	0	0
HED	DDR	6	.191	.430	.239	2	0	0
HED	IDRiD	7	.366	.639	.274	2	0	0
HRNet	DDR	4	.452	.483	.031	3	0	0
HRNet	IDRiD	4	.649	.689	.040	3	0	0
HRNet+OCR	DDR	2	.452	.474	.022	0	0	0
HRNet+OCR	IDRiD	2	.649	.660	.011	0	0	0
L-Seg	DDR	13	.321	.321	.000	0	0	0
L-Seg	IDRiD	14	.399	.661	.262	0	0	0
M2MRF	DDR	9	.489	.499	.010	0	0	0
M2MRF	IDRiD	9	.583	.672	.089	0	0	0
MedLabs	IDRiD	3	.440	.440	.000	0	0	0
MedSAM	DDR	2	.193	.193	.000	0	0	0
MedSAM	IDRiD	2	.266	.266	.000	0	0	0
PMCNet	DDR	4	.364	.364	.000	0	0	0
PMCNet	IDRiD	3	.681	.681	.000	0	0	0
PSPNet	DDR	4	.177	.392	.215	3	0	0
PSPNet	IDRiD	4	.337	.587	.250	3	0	0
RILBP-YNet	IDRiD	3	.659	.659	.000	0	0	0
RTNet	DDR	4	.336	.336	.000	0	0	0
RTNet	IDRiD	3	.421	.708	.287	0	0	0
SAA	DDR	4	.366	.410	.044	0	0	0
SAA	IDRiD	5	.648	.674	.026	0	0	0
SAM	DDR	2	.003	.003	.000	0	0	0
SAM	IDRiD	2	.006	.006	.000	0	0	0
SAM-Adapter	DDR	2	.299	.299	.000	0	0	0
SAM-Adapter	IDRiD	2	.346	.346	.000	0	0	0
SAMed	DDR	2	.300	.300	.000	0	0	0
SAMed	IDRiD	2	.354	.354	.000	0	0	0
SDNU	IDRiD	3	.477	.477	.000	0	0	0
SOONER	IDRiD	3	.554	.554	.000	0	0	0
Swin (UPerNet)	DDR	5	.468	.486	.018	2	0	0
Swin (UPerNet)	IDRiD	4	.649	.690	.042	2	0	0
Twins-SVT-B	DDR	2	.461	.461	.000	1	0	0
Twins-SVT-B	IDRiD	2	.638	.638	.000	1	0	0
U-Net	DDR	7	.190	.429	.239	2	0	0
U-Net	IDRiD	6	.347	.652	.305	2	0	0
U-Net++	DDR	4	.192	.409	.217	3	0	0
U-Net++	IDRiD	3	.391	.572	.181	3	0	0
VRT	IDRiD	8	.647	.647	.000	0	0	0
iFLYTEK-MIG	IDRiD	7	.648	.648	.000	0	0	0

Values as printed, normalised to [0, 1]. The full extraction, one row per paper, baseline, dataset, class and metric, is literature/meta/reported_results.tsv.

Supplementary Table 3 Training amount, batch and selection of every main-table run (seed 0). Steps are the shared pixel budget expressed in the model's patch size (Methods); the loader batch is the largest that fits a 40 GB card and is accumulated to the recipe's effective batch; the selected checkpoint is the validation mAUPR maximum over 20 evaluations. nnU-Net follows its own 1,000-epoch schedule and planning.

Model	Source	Steps	Loader batch	Accum.	Eff. batch	Patch	LR	Best val. mAUPR	GPU h
U-Net	IDRiD	21,000	4	1	4	960×960	0.0001	0.582	1.5
U-Net	DDR	83,600	4	1	4	960×960	0.0001	0.556	5.4
DeepLabv3	IDRiD	6,000	1	16	16	960×960	0.007	0.428	3.6
DeepLabv3	DDR	21,200	1	16	16	960×960	0.007	0.436	12.8
HRNet-W48	IDRiD	7,000	2	6	12	960×960	0.01	0.618	2.6
HRNet-W48	DDR	28,000	2	6	12	960×960	0.01	0.617	10.1
Swin-Unet	IDRiD	63,000	24	1	24	224×224	0.01	0.644	2.6
Swin-Unet	DDR	255,600	24	1	24	224×224	0.01	0.620	10.4
FCT	IDRiD	129,000	4	1	4	384×384	0.001	0.257	19.7
FCT	DDR	521,600	4	1	4	384×384	0.001	0.175	72.0
nnU-Net	IDRiD	250,000	2	1	2	1024×1536	0.01	–[n]	7.2
nnU-Net	DDR	250,000	2	1	2	1024×1536	0.01	–[n]	7.2
nnU-Net ResEnc-M	IDRiD	250,000	13	1	13	384×640	0.01	–[n]	12.8
nnU-Net ResEnc-M	DDR	250,000	12	1	12	512×512	0.01	–[n]	11.4
M2MRF	IDRiD	21,000	4	1	4	960×960	0.01	0.541	3.0
M2MRF	DDR	83,600	4	1	4	960×960	0.01	0.558	11.6
HACDR-Net	IDRiD	83,000	1	1	1	960×960	6e-05	0.518	5.9
HACDR-Net	DDR	334,000	1	1	1	960×960	6e-05	0.383	27.8
HRDecoder	IDRiD	21,000	4	1	4	960×960	0.01	0.562	1.5
HRDecoder	DDR	83,600	4	1	4	960×960	0.01	0.539	5.9
PMCNet	IDRiD	83,000	1	1	1	960×960	0.0001	0.544	1.8
PMCNet	DDR	334,000	1	1	1	960×960	0.0001	0.512	7.1
WSRFNet	IDRiD	21,000	4	1	4	960×960	0.01	0.511	2.7
WSRFNet	DDR	83,600	4	1	4	960×960	0.01	0.524	10.5
CLC-Net	IDRiD	73,000	16	1	16	512×512	0.0003	0.580	9.9
CLC-Net	DDR	293,600	16	1	16	512×512	0.0003	0.539	39.2
TC-Net	IDRiD	73,000	4	1	4	512×512	0.0005	0.570	6.0
TC-Net	DDR	293,600	4	1	4	512×512	0.0005	0.472	24.7
SAMed	IDRiD	13,000	24	1	24	512×512	0.005	0.637	2.1
SAMed	DDR	49,200	24	1	24	512×512	0.005	0.632	7.4
SAM-Adapter	IDRiD	37,000	2	1	2	1024×1024	0.0002	0.617	6.5
SAM-Adapter	DDR	146,800	2	1	2	1024×1024	0.0002	0.589	26.3
SAM2-UNet	IDRiD	7,000	4	3	12	960×960	0.001	0.629	4.1
SAM2-UNet	DDR	28,000	4	3	12	960×960	0.001	0.540	16.2
MedSAM (full FT)	IDRiD	37,000	2	1	2	1024×1024	0.0001	0.631	2.8
MedSAM (full FT)	DDR	146,800	2	1	2	1024×1024	0.0001	0.509	11.1
RETFound + UPerNet	IDRiD	63,000	24	1	24	224×224	0.00046875	0.645	6.1
RETFound + UPerNet	DDR	255,600	24	1	24	224×224	0.00046875	0.643	24.2

Every run's full recipe.json (deviations from the source recipe included) is in the repository's run records. nnU-Net: patch and batch from its own plan for each dataset (runs/NN_*/console.log), 1,000 epochs of 250 steps, SGD at 0.01 with polynomial decay; [n] its checkpoint is nnU-Net's own (the final checkpoint of its schedule), so no validation mAUPR maximum applies.

Supplementary Table 4 Per-class AUPR of every main-table model on every test set (seed 0), with 95% bootstrap intervals. Exact pooled AUPR inside the field of view; the interval is the percentile interval of 2,000 bootstrap replicates over test images. Both training sources; the internal test is the row whose test set is the training source.

Model	Test set	MA	95% CI	HE	95% CI	EX	95% CI	SE	95% CI	mAUPR
Trained on IDRiD										
U-Net	IDRiD	.406	.33–.44	.622	.57–.67	.896	.83–.92	.707	.45–.74	.658
U-Net	DDR	.110	.08–.14	.377	.31–.41	.477	.39–.53	.199	.05–.23	.291
U-Net	TJDR std	.255	.13–.30	.626	.51–.71	.560	.43–.65	.235	.05–.31	.419
U-Net	TJDR UWF	.079	.04–.13	.411	.24–.53	.164	.08–.27	.180	.07–.31	.208
U-Net	MAPLES	.264	.16–.29	.328	.16–.48	.617	.32–.73	.035	.00–.12	.311
U-Net	FGADR	.010	.01–.01	.200	.16–.24	.255	.21–.29	.107	.07–.13	.143
U-Net	Retinal-L.	.044	.03–.04	.186	.13–.18	.208	.13–.21	.104	.06–.12	.136
U-Net	DiaRetDB1	.036	.02–.06	.269	.17–.33	.285	.18–.28	.112	.03–.19	.176
DeepLabv3	IDRiD	.090	.07–.12	.643	.54–.69	.734	.62–.79	.662	.49–.76	.532
DeepLabv3	DDR	.042	.03–.06	.327	.26–.40	.511	.43–.59	.149	.07–.24	.257
DeepLabv3	TJDR std	.133	.07–.21	.605	.49–.69	.585	.32–.75	.307	.12–.52	.407
DeepLabv3	TJDR UWF	.058	.02–.14	.517	.31–.63	.320	.18–.47	.213	.10–.39	.277
DeepLabv3	MAPLES	.160	.12–.21	.237	.08–.43	.661	.41–.77	.087	.01–.21	.286
DeepLabv3	FGADR	.016	.01–.02	.334	.30–.37	.335	.27–.40	.183	.14–.23	.217
DeepLabv3	Retinal-L.	.093	.08–.10	.349	.31–.38	.406	.31–.50	.195	.15–.24	.261
DeepLabv3	DiaRetDB1	.010	.01–.02	.510	.41–.59	.559	.46–.61	.209	.03–.41	.322
HRNet-W48	IDRiD	.446	.41–.48	.684	.63–.72	.884	.82–.91	.783	.59–.87	.699
HRNet-W48	DDR	.146	.11–.19	.438	.38–.51	.507	.42–.58	.254	.12–.38	.336
HRNet-W48	TJDR std	.323	.20–.42	.706	.60–.79	.561	.40–.69	.283	.11–.52	.468
HRNet-W48	TJDR UWF	.111	.05–.20	.480	.28–.64	.132	.07–.25	.296	.13–.49	.255
HRNet-W48	MAPLES	.394	.32–.47	.492	.30–.62	.646	.38–.78	.078	.01–.24	.403
HRNet-W48	FGADR	.025	.02–.03	.375	.33–.42	.362	.30–.42	.234	.19–.29	.249
HRNet-W48	Retinal-L.	.070	.06–.08	.259	.22–.30	.262	.20–.32	.198	.14–.25	.197
HRNet-W48	DiaRetDB1	.039	.02–.06	.456	.36–.54	.382	.27–.45	.212	.06–.34	.272
Swin-Unet	IDRiD	.478	.44–.51	.709	.65–.75	.894	.81–.93	.685	.42–.82	.692
Swin-Unet	DDR	.132	.10–.17	.400	.33–.48	.559	.49–.62	.216	.10–.34	.327
Swin-Unet	TJDR std	.341	.23–.43	.495	.38–.59	.711	.60–.80	.347	.16–.54	.474
Swin-Unet	TJDR UWF	.090	.04–.16	.471	.29–.57	.145	.07–.27	.189	.07–.35	.224
Swin-Unet	MAPLES	.378	.31–.44	.454	.26–.60	.662	.38–.79	.120	.03–.24	.404
Swin-Unet	FGADR	.018	.01–.02	.300	.26–.34	.303	.25–.35	.173	.12–.23	.198
Swin-Unet	Retinal-L.	.083	.07–.09	.289	.24–.32	.359	.27–.42	.197	.15–.24	.232
Swin-Unet	DiaRetDB1	.058	.03–.08	.459	.37–.53	.458	.34–.51	.237	.07–.38	.303
FCT	IDRiD	.077	.06–.10	.163	.10–.23	.684	.57–.74	.110	.03–.26	.259
FCT	DDR	.008	.00–.01	.067	.05–.09	.224	.13–.32	.004	.00–.01	.076
FCT	TJDR std	.000	.00–.00	.006	.00–.01	.004	.00–.01	.004	.00–.01	.004
FCT	TJDR UWF	.001	.00–.00	.020	.01–.04	.021	.01–.05	.006	.00–.01	.012
FCT	MAPLES	.020	.01–.03	.018	.00–.05	.465	.15–.64	.003	.00–.01	.126
FCT	FGADR	.005	.00–.01	.061	.05–.07	.112	.08–.15	.009	.01–.01	.046
FCT	Retinal-L.	.011	.01–.01	.037	.03–.04	.049	.03–.07	.007	.01–.01	.026
FCT	DiaRetDB1	.004	.00–.01	.102	.04–.17	.275	.15–.37	.002	.00–.01	.096
nnU-Net	IDRiD	.499	.42–.51	.671	.61–.72	.876	.79–.91	.641	.40–.79	.671
nnU-Net	DDR	.122	.08–.14	.384	.24–.45	.481	.37–.54	.200	.09–.27	.297
nnU-Net	TJDR std	.210	.13–.25	.615	.48–.70	.490	.32–.60	.421	.21–.64	.434
nnU-Net	TJDR UWF	.088	.03–.15	.419	.20–.59	.169	.08–.25	.144	.06–.26	.205
nnU-Net	MAPLES	.314	.19–.29	.455	.26–.56	.682	.37–.78	.151	.05–.28	.401
nnU-Net	FGADR	.018	.01–.02	.370	.29–.38	.360	.29–.38	.201	.14–.24	.237
nnU-Net	Retinal-L.	.055	.04–.04	.283	.19–.23	.289	.18–.27	.146	.09–.15	.193
nnU-Net	DiaRetDB1	.035	.02–.05	.437	.30–.45	.428	.26–.36	.171	.05–.29	.268

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Model	Test set	MA	95% CI	HE	95% CI	EX	95% CI	SE	95% CI	mAUPR
nnU-Net ResEnc-M	IDRiD	.449	.36–.44	.663	.58–.68	.878	.79–.90	.533	.29–.68	.631
nnU-Net ResEnc-M	DDR	.113	.08–.12	.337	.21–.38	.476	.36–.53	.165	.07–.24	.273
nnU-Net ResEnc-M	TJDR std	.217	.12–.25	.555	.41–.61	.526	.36–.61	.385	.15–.58	.421
nnU-Net ResEnc-M	TJDR UWF	.071	.02–.13	.421	.18–.60	.134	.05–.22	.143	.06–.26	.192
nnU-Net ResEnc-M	MAPLES	.269	.17–.26	.472	.25–.57	.677	.34–.74	.017	.00–.08	.359
nnU-Net ResEnc-M	FGADR	.017	.01–.02	.322	.24–.33	.332	.26–.36	.148	.10–.18	.205
nnU-Net ResEnc-M	Retinal-L.	.043	.03–.04	.192	.14–.17	.227	.15–.23	.086	.05–.11	.137
nnU-Net ResEnc-M	DiaRetDB1	.035	.02–.04	.319	.20–.33	.321	.21–.29	.232	.05–.31	.227
M2MRF	IDRiD	.443	.38–.46	.615	.55–.64	.868	.77–.89	.675	.47–.76	.650
M2MRF	DDR	.106	.08–.13	.294	.20–.36	.410	.31–.46	.127	.06–.20	.234
M2MRF	TJDR std	.259	.17–.31	.482	.33–.56	.608	.49–.69	.286	.07–.48	.409
M2MRF	TJDR UWF	.038	.01–.07	.360	.21–.44	.035	.01–.10	.205	.08–.29	.160
M2MRF	MAPLES	.253	.18–.29	.400	.21–.51	.547	.32–.65	.044	.00–.10	.311
M2MRF	FGADR	.007	.01–.01	.082	.06–.09	.172	.11–.22	.083	.05–.11	.086
M2MRF	Retinal-L.	.044	.04–.05	.172	.12–.17	.185	.11–.21	.089	.05–.11	.122
M2MRF	DiaRetDB1	.038	.02–.05	.354	.23–.38	.233	.16–.23	.165	.03–.28	.198
HACDR-Net	IDRiD	.359	.31–.40	.502	.41–.61	.827	.74–.88	.560	.34–.68	.562
HACDR-Net	DDR	.091	.05–.14	.244	.17–.31	.266	.15–.45	.060	.01–.17	.165
HACDR-Net	TJDR std	.378	.19–.52	.475	.33–.59	.641	.55–.72	.529	.16–.76	.506
HACDR-Net	TJDR UWF	.071	.03–.19	.267	.15–.39	.103	.03–.23	.058	.01–.16	.125
HACDR-Net	MAPLES	.280	.21–.35	.177	.04–.41	.579	.27–.73	.035	.00–.15	.268
HACDR-Net	FGADR	.015	.01–.02	.069	.05–.09	.044	.03–.06	.008	.00–.02	.034
HACDR-Net	Retinal-L.	.040	.03–.04	.064	.04–.08	.183	.12–.21	.022	.01–.04	.077
HACDR-Net	DiaRetDB1	.044	.02–.07	.290	.17–.42	.383	.22–.43	.072	.01–.17	.197
HRDecoder	IDRiD	.410	.36–.43	.629	.56–.66	.823	.70–.85	.728	.48–.81	.647
HRDecoder	DDR	.110	.08–.13	.388	.30–.44	.530	.44–.58	.168	.06–.22	.299
HRDecoder	TJDR std	.236	.14–.29	.423	.26–.54	.593	.43–.67	.318	.13–.46	.393
HRDecoder	TJDR UWF	.107	.04–.18	.362	.15–.46	.157	.08–.23	.131	.05–.31	.189
HRDecoder	MAPLES	.326	.24–.37	.464	.26–.59	.597	.33–.70	.121	.03–.23	.377
HRDecoder	FGADR	.020	.02–.02	.323	.27–.35	.293	.22–.34	.179	.13–.22	.204
HRDecoder	Retinal-L.	.053	.04–.05	.180	.14–.18	.182	.12–.20	.116	.07–.14	.133
HRDecoder	DiaRetDB1	.052	.03–.07	.340	.24–.37	.218	.15–.23	.217	.06–.33	.207
PMCNet	IDRiD	.418	.36–.47	.610	.56–.67	.883	.80–.92	.698	.41–.82	.652
PMCNet	DDR	.133	.10–.17	.324	.22–.43	.464	.34–.57	.135	.04–.28	.264
PMCNet	TJDR std	.372	.22–.48	.600	.44–.71	.506	.39–.62	.248	.05–.43	.432
PMCNet	TJDR UWF	.125	.04–.21	.431	.23–.58	.210	.11–.35	.254	.13–.40	.255
PMCNet	MAPLES	.344	.25–.43	.334	.14–.50	.616	.35–.75	.052	.00–.16	.337
PMCNet	FGADR	.023	.02–.03	.335	.29–.38	.355	.31–.40	.133	.09–.18	.212
PMCNet	Retinal-L.	.076	.06–.07	.213	.16–.24	.305	.21–.32	.129	.08–.16	.181
PMCNet	DiaRetDB1	.033	.02–.05	.325	.22–.41	.396	.27–.40	.153	.01–.27	.227
WSRFNet	IDRiD	.388	.33–.40	.588	.51–.61	.849	.70–.88	.502	.18–.65	.582
WSRFNet	DDR	.087	.06–.10	.287	.19–.34	.473	.38–.49	.186	.09–.27	.258
WSRFNet	TJDR std	.198	.12–.24	.426	.27–.51	.539	.40–.59	.220	.06–.43	.346
WSRFNet	TJDR UWF	.061	.03–.09	.379	.22–.48	.116	.04–.20	.230	.09–.32	.196
WSRFNet	MAPLES	.245	.18–.27	.411	.22–.51	.544	.29–.62	.032	.00–.10	.308
WSRFNet	FGADR	.010	.01–.01	.300	.25–.31	.314	.25–.34	.162	.11–.19	.197
WSRFNet	Retinal-L.	.043	.04–.04	.141	.11–.15	.172	.11–.20	.148	.10–.16	.126
WSRFNet	DiaRetDB1	.038	.02–.05	.264	.19–.31	.221	.16–.23	.120	.02–.20	.161
CLC-Net	IDRiD	.437	.37–.46	.683	.59–.71	.889	.81–.92	.634	.40–.73	.661
CLC-Net	DDR	.115	.08–.14	.439	.36–.48	.539	.45–.57	.154	.06–.23	.312
CLC-Net	TJDR std	.295	.16–.38	.631	.50–.71	.618	.45–.73	.292	.12–.43	.459
CLC-Net	TJDR UWF	.091	.03–.16	.466	.25–.57	.170	.08–.26	.105	.04–.18	.208
CLC-Net	MAPLES	.336	.23–.37	.477	.28–.59	.689	.41–.78	.065	.01–.16	.392

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continued

Model	Test set	MA	95% CI	HE	95% CI	EX	95% CI	SE	95% CI	mAUPR
CLC-Net	FGADR	.007	.01–.01	.318	.27–.35	.288	.24–.31	.086	.05–.11	.175
CLC-Net	Retinal-L.	.060	.04–.05	.246	.19–.26	.348	.20–.32	.105	.07–.12	.190
CLC-Net	DiaRetDB1	.044	.02–.06	.400	.28–.48	.444	.25–.36	.140	.03–.23	.257
TC-Net	IDRiD	.355	.31–.40	.650	.59–.70	.854	.78–.89	.754	.57–.85	.653
TC-Net	DDR	.119	.09–.15	.404	.32–.49	.572	.50–.63	.153	.07–.25	.312
TC-Net	TJDR std	.233	.16–.31	.605	.50–.69	.691	.58–.77	.398	.15–.61	.482
TC-Net	TJDR UWF	.124	.07–.22	.489	.27–.61	.017	.01–.04	.223	.12–.34	.213
TC-Net	MAPLES	.372	.29–.46	.453	.27–.58	.682	.39–.81	.143	.03–.30	.412
TC-Net	FGADR	.022	.02–.02	.366	.32–.41	.318	.26–.37	.144	.11–.19	.212
TC-Net	Retinal-L.	.080	.07–.08	.270	.22–.30	.237	.16–.29	.114	.08–.15	.175
TC-Net	DiaRetDB1	.039	.02–.06	.433	.34–.51	.426	.29–.48	.218	.09–.32	.279
SAMed	IDRiD	.449	.40–.50	.736	.67–.78	.892	.82–.92	.790	.61–.88	.717
SAMed	DDR	.128	.10–.16	.443	.37–.52	.609	.54–.67	.339	.18–.48	.380
SAMed	TJDR std	.301	.20–.38	.669	.53–.76	.706	.61–.79	.559	.25–.74	.559
SAMed	TJDR UWF	.115	.04–.19	.470	.24–.66	.226	.13–.36	.158	.05–.33	.242
SAMed	MAPLES	.351	.26–.44	.471	.30–.60	.684	.40–.80	.136	.04–.30	.410
SAMed	FGADR	.025	.02–.03	.418	.38–.46	.409	.36–.46	.282	.23–.34	.283
SAMed	Retinal-L.	.083	.07–.09	.354	.31–.38	.362	.28–.42	.219	.16–.26	.255
SAMed	DiaRetDB1	.055	.03–.08	.473	.39–.54	.455	.35–.50	.276	.08–.45	.315
SAM-Adapter	IDRiD	.451	.39–.46	.740	.66–.78	.882	.79–.92	.750	.54–.82	.706
SAM-Adapter	DDR	.117	.08–.14	.403	.30–.46	.560	.47–.59	.157	.08–.23	.309
SAM-Adapter	TJDR std	.256	.14–.33	.581	.43–.69	.563	.40–.63	.351	.16–.55	.438
SAM-Adapter	TJDR UWF	.067	.02–.10	.447	.21–.58	.125	.05–.21	.196	.09–.32	.209
SAM-Adapter	MAPLES	.292	.19–.32	.395	.21–.50	.605	.31–.71	.144	.05–.27	.359
SAM-Adapter	FGADR	.018	.01–.02	.355	.29–.39	.343	.27–.39	.166	.12–.22	.221
SAM-Adapter	Retinal-L.	.046	.04–.04	.246	.17–.23	.317	.17–.32	.142	.10–.16	.188
SAM-Adapter	DiaRetDB1	.037	.02–.06	.449	.30–.49	.365	.19–.32	.130	.03–.26	.245
SAM2-UNet	IDRiD	.439	.37–.45	.670	.59–.72	.871	.78–.90	.678	.43–.77	.664
SAM2-UNet	DDR	.112	.08–.13	.448	.38–.50	.516	.42–.56	.239	.05–.17	.329
SAM2-UNet	TJDR std	.289	.16–.35	.680	.58–.75	.643	.53–.69	.510	.14–.52	.530
SAM2-UNet	TJDR UWF	.091	.03–.14	.482	.30–.62	.121	.04–.23	.230	.06–.31	.231
SAM2-UNet	MAPLES	.319	.23–.37	.404	.21–.53	.644	.38–.75	.097	.01–.21	.366
SAM2-UNet	FGADR	.016	.01–.02	.378	.32–.40	.377	.30–.40	.242	.16–.24	.253
SAM2-UNet	Retinal-L.	.063	.05–.06	.249	.18–.25	.274	.18–.29	.159	.08–.15	.186
SAM2-UNet	DiaRetDB1	.037	.02–.05	.327	.24–.39	.319	.23–.30	.197	.04–.30	.220
MedSAM (full FT)	IDRiD	.373	.33–.41	.665	.57–.73	.855	.77–.90	.674	.43–.83	.642
MedSAM (full FT)	DDR	.144	.11–.18	.400	.35–.46	.555	.49–.62	.155	.08–.25	.314
MedSAM (full FT)	TJDR std	.224	.17–.27	.593	.42–.71	.600	.47–.70	.395	.19–.57	.453
MedSAM (full FT)	TJDR UWF	.119	.05–.24	.443	.24–.57	.213	.12–.34	.403	.25–.55	.295
MedSAM (full FT)	MAPLES	.331	.27–.39	.364	.17–.52	.650	.38–.77	.132	.05–.25	.369
MedSAM (full FT)	FGADR	.024	.02–.03	.367	.33–.41	.394	.35–.44	.254	.20–.31	.260
MedSAM (full FT)	Retinal-L.	.078	.07–.08	.260	.22–.29	.394	.33–.45	.125	.10–.15	.214
MedSAM (full FT)	DiaRetDB1	.043	.02–.07	.417	.30–.52	.535	.42–.60	.131	.03–.24	.282
RETFound + UPerNet	IDRiD	.484	.45–.52	.718	.66–.75	.895	.82–.93	.734	.50–.85	.708
RETFound + UPerNet	DDR	.152	.12–.19	.413	.32–.50	.572	.50–.63	.278	.13–.43	.354
RETFound + UPerNet	TJDR std	.393	.25–.49	.582	.46–.67	.776	.68–.86	.070	.02–.15	.455
RETFound + UPerNet	TJDR UWF	.099	.04–.18	.494	.28–.63	.106	.05–.27	.311	.15–.51	.253
RETFound + UPerNet	MAPLES	.375	.29–.45	.503	.30–.64	.680	.38–.81	.092	.03–.23	.413
RETFound + UPerNet	FGADR	.023	.02–.03	.332	.29–.38	.357	.29–.42	.215	.16–.28	.232
RETFound + UPerNet	Retinal-L.	.081	.06–.08	.288	.23–.32	.299	.22–.36	.156	.11–.20	.206
RETFound + UPerNet	DiaRetDB1	.039	.02–.06	.459	.34–.56	.474	.29–.55	.257	.07–.40	.307
Trained on DDR										
U-Net	IDRiD	.364	.29–.36	.541	.43–.57	.800	.66–.78	.483	.26–.56	.547

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Model	Test set	MA	95% CI	HE	95% CI	EX	95% CI	SE	95% CI	mAUPR
U-Net	DDR	.141	.09–.16	.410	.28–.48	.553	.42–.59	.410	.23–.49	.378
U-Net	TJDR std	.296	.16–.34	.676	.55–.75	.614	.42–.65	.392	.13–.49	.494
U-Net	TJDR UWF	.077	.03–.21	.262	.11–.39	.254	.15–.32	.067	.01–.13	.165
U-Net	MAPLES	.222	.13–.25	.462	.24–.58	.509	.24–.50	.125	.01–.21	.329
U-Net	FGADR	.013	.01–.01	.214	.18–.24	.292	.23–.33	.162	.11–.19	.170
U-Net	Retinal-L.	.039	.03–.04	.165	.10–.16	.167	.10–.16	.197	.13–.21	.142
U-Net	DiaRetDB1	.039	.02–.06	.314	.19–.34	.268	.12–.20	.253	.11–.31	.219
DeepLabv3	IDRiD	.052	.04–.07	.640	.49–.72	.708	.58–.77	.630	.40–.75	.507
DeepLabv3	DDR	.036	.02–.05	.447	.30–.61	.614	.54–.68	.387	.25–.50	.371
DeepLabv3	TJDR std	.172	.08–.28	.649	.51–.75	.724	.64–.76	.411	.15–.61	.489
DeepLabv3	TJDR UWF	.071	.03–.10	.066	.02–.20	.296	.17–.44	.058	.02–.18	.123
DeepLabv3	MAPLES	.143	.10–.19	.452	.24–.58	.641	.41–.73	.108	.01–.24	.336
DeepLabv3	FGADR	.011	.01–.01	.339	.30–.38	.356	.30–.41	.224	.18–.27	.232
DeepLabv3	Retinal-L.	.074	.06–.08	.316	.25–.37	.440	.32–.52	.253	.19–.31	.271
DeepLabv3	DiaRetDB1	.008	.00–.01	.495	.40–.57	.545	.44–.59	.241	.03–.40	.322
HRNet-W48	IDRiD	.372	.34–.40	.694	.58–.76	.834	.75–.88	.679	.42–.79	.645
HRNet-W48	DDR	.155	.12–.20	.477	.39–.59	.590	.52–.67	.550	.42–.65	.443
HRNet-W48	TJDR std	.412	.24–.53	.788	.71–.85	.713	.63–.76	.581	.25–.76	.624
HRNet-W48	TJDR UWF	.116	.06–.24	.120	.04–.29	.248	.15–.36	.107	.02–.27	.148
HRNet-W48	MAPLES	.383	.31–.45	.500	.28–.63	.701	.48–.79	.220	.05–.37	.451
HRNet-W48	FGADR	.026	.02–.03	.399	.35–.45	.394	.34–.45	.255	.20–.32	.269
HRNet-W48	Retinal-L.	.083	.07–.09	.251	.20–.31	.339	.27–.40	.332	.28–.38	.251
HRNet-W48	DiaRetDB1	.034	.02–.05	.437	.34–.53	.432	.32–.52	.362	.21–.46	.316
Swin-Unet	IDRiD	.420	.38–.46	.625	.52–.69	.853	.76–.89	.615	.41–.73	.628
Swin-Unet	DDR	.174	.13–.22	.479	.39–.58	.650	.59–.71	.424	.29–.53	.432
Swin-Unet	TJDR std	.370	.24–.45	.676	.56–.76	.696	.53–.79	.438	.13–.63	.545
Swin-Unet	TJDR UWF	.107	.05–.22	.336	.19–.47	.271	.18–.37	.044	.01–.14	.189
Swin-Unet	MAPLES	.395	.32–.46	.509	.31–.64	.726	.50–.81	.200	.06–.35	.457
Swin-Unet	FGADR	.021	.02–.02	.349	.31–.39	.382	.33–.44	.216	.17–.27	.242
Swin-Unet	Retinal-L.	.113	.09–.12	.254	.20–.30	.418	.33–.48	.242	.19–.29	.257
Swin-Unet	DiaRetDB1	.048	.03–.07	.381	.28–.47	.496	.40–.55	.255	.14–.36	.295
FCT	IDRiD	.005	.00–.01	.153	.10–.21	.679	.56–.74	.046	.01–.13	.221
FCT	DDR	.001	.00–.00	.163	.11–.22	.524	.46–.58	.014	.01–.02	.176
FCT	TJDR std	.001	.00–.00	.099	.05–.15	.596	.43–.68	.040	.01–.11	.184
FCT	TJDR UWF	.001	.00–.00	.055	.02–.10	.135	.06–.24	.009	.00–.02	.050
FCT	MAPLES	.004	.00–.01	.030	.01–.08	.510	.21–.66	.007	.00–.02	.138
FCT	FGADR	.004	.00–.00	.065	.05–.08	.182	.14–.23	.015	.01–.02	.067
FCT	Retinal-L.	.010	.01–.01	.057	.04–.07	.210	.14–.28	.018	.01–.02	.074
FCT	DiaRetDB1	.000	.00–.00	.087	.02–.17	.270	.16–.34	.009	.00–.02	.092
nnU-Net	IDRiD	.334	.24–.34	.637	.52–.71	.810	.71–.86	.623	.38–.71	.601
nnU-Net	DDR	.154	.11–.17	.501	.39–.57	.614	.51–.66	.458	.30–.56	.432
nnU-Net	TJDR std	.309	.17–.36	.705	.61–.75	.635	.45–.71	.425	.15–.61	.518
nnU-Net	TJDR UWF	.111	.03–.21	.374	.18–.54	.157	.07–.26	.142	.06–.27	.196
nnU-Net	MAPLES	.372	.18–.29	.493	.27–.58	.717	.38–.75	.196	.06–.29	.445
nnU-Net	FGADR	.020	.01–.02	.396	.33–.41	.389	.31–.42	.232	.18–.27	.259
nnU-Net	Retinal-L.	.067	.04–.05	.232	.14–.19	.319	.17–.28	.224	.17–.23	.211
nnU-Net	DiaRetDB1	.065	.04–.09	.462	.31–.48	.476	.26–.39	.295	.10–.35	.325
nnU-Net ResEnc-M	IDRiD	.349	.26–.32	.664	.54–.72	.832	.73–.87	.603	.36–.67	.612
nnU-Net ResEnc-M	DDR	.147	.10–.16	.481	.37–.55	.653	.56–.70	.432	.28–.52	.428
nnU-Net ResEnc-M	TJDR std	.320	.19–.34	.751	.63–.81	.673	.48–.75	.538	.19–.69	.571
nnU-Net ResEnc-M	TJDR UWF	.115	.03–.21	.317	.17–.44	.132	.06–.24	.112	.02–.23	.169
nnU-Net ResEnc-M	MAPLES	.341	.16–.29	.469	.26–.58	.762	.42–.78	.293	.12–.37	.466
nnU-Net ResEnc-M	FGADR	.018	.01–.02	.409	.33–.41	.401	.31–.42	.242	.17–.27	.268

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Model	Test set	MA	95% CI	HE	95% CI	EX	95% CI	SE	95% CI	mAUPR
nnU-Net ResEnc-M	Retinal-L.	.062	.03–.04	.269	.13–.21	.355	.20–.29	.257	.18–.24	.236
nnU-Net ResEnc-M	DiaRetDB1	.062	.03–.08	.450	.28–.44	.475	.27–.37	.261	.06–.32	.312
M2MRF	IDRiD	.392	.34–.40	.555	.45–.56	.776	.68–.78	.563	.35–.66	.571
M2MRF	DDR	.148	.10–.18	.402	.27–.48	.547	.46–.59	.422	.25–.49	.380
M2MRF	TJDR std	.299	.19–.35	.696	.58–.75	.618	.50–.68	.412	.14–.55	.506
M2MRF	TJDR UWF	.093	.04–.17	.403	.22–.51	.209	.11–.30	.037	.01–.10	.186
M2MRF	MAPLES	.266	.17–.30	.437	.24–.55	.454	.30–.50	.138	.02–.25	.324
M2MRF	FGADR	.019	.01–.02	.310	.24–.33	.340	.27–.38	.181	.13–.22	.212
M2MRF	Retinal-L.	.050	.04–.05	.183	.13–.17	.174	.12–.18	.195	.13–.22	.150
M2MRF	DiaRetDB1	.049	.02–.07	.218	.15–.26	.177	.13–.18	.247	.14–.30	.173
HACDR-Net	IDRiD	.225	.18–.26	.365	.28–.47	.789	.67–.83	.199	.04–.44	.395
HACDR-Net	DDR	.114	.08–.16	.309	.22–.40	.485	.37–.58	.168	.08–.28	.269
HACDR-Net	TJDR std	.356	.21–.46	.632	.54–.71	.553	.45–.64	.196	.03–.39	.434
HACDR-Net	TJDR UWF	.095	.03–.20	.385	.20–.50	.204	.11–.32	.130	.04–.24	.203
HACDR-Net	MAPLES	.224	.15–.29	.378	.17–.54	.512	.22–.63	.093	.01–.22	.302
HACDR-Net	FGADR	.021	.02–.02	.186	.15–.22	.134	.10–.17	.049	.03–.07	.098
HACDR-Net	Retinal-L.	.058	.04–.06	.163	.12–.19	.166	.11–.19	.138	.10–.17	.131
HACDR-Net	DiaRetDB1	.016	.01–.03	.228	.12–.32	.319	.15–.34	.101	.02–.19	.166
HRDecoder	IDRiD	.357	.31–.36	.547	.41–.57	.714	.60–.73	.539	.35–.60	.539
HRDecoder	DDR	.124	.08–.16	.318	.20–.40	.496	.40–.56	.245	.11–.33	.296
HRDecoder	TJDR std	.334	.19–.41	.689	.59–.75	.614	.51–.67	.358	.12–.51	.499
HRDecoder	TJDR UWF	.119	.05–.18	.368	.17–.53	.180	.09–.27	.124	.05–.23	.198
HRDecoder	MAPLES	.231	.14–.26	.407	.23–.50	.427	.22–.48	.127	.03–.22	.298
HRDecoder	FGADR	.014	.01–.02	.267	.21–.29	.309	.25–.35	.183	.14–.22	.193
HRDecoder	Retinal-L.	.043	.03–.04	.126	.08–.14	.171	.11–.17	.168	.13–.19	.127
HRDecoder	DiaRetDB1	.045	.02–.06	.275	.17–.32	.234	.13–.19	.203	.08–.29	.189
PMCNet	IDRiD	.348	.31–.38	.458	.37–.51	.758	.68–.81	.524	.20–.71	.522
PMCNet	DDR	.157	.12–.20	.346	.23–.46	.482	.37–.58	.262	.14–.38	.312
PMCNet	TJDR std	.406	.25–.50	.639	.54–.72	.476	.33–.60	.308	.06–.49	.457
PMCNet	TJDR UWF	.085	.03–.19	.427	.23–.55	.217	.12–.32	.091	.03–.19	.205
PMCNet	MAPLES	.371	.28–.44	.444	.24–.59	.627	.34–.74	.131	.00–.28	.393
PMCNet	FGADR	.021	.02–.02	.323	.27–.37	.339	.29–.37	.162	.11–.21	.211
PMCNet	Retinal-L.	.105	.08–.10	.236	.19–.25	.272	.18–.26	.128	.09–.16	.185
PMCNet	DiaRetDB1	.036	.02–.06	.312	.21–.40	.366	.23–.38	.266	.04–.42	.245
WSRFNet	IDRiD	.319	.24–.32	.489	.36–.51	.673	.53–.72	.384	.17–.49	.466
WSRFNet	DDR	.088	.05–.11	.292	.20–.37	.422	.31–.48	.135	.05–.21	.234
WSRFNet	TJDR std	.173	.08–.26	.588	.43–.64	.433	.23–.51	.114	.01–.20	.327
WSRFNet	TJDR UWF	.038	.01–.18	.323	.16–.39	.155	.06–.24	.024	.00–.08	.135
WSRFNet	MAPLES	.264	.17–.29	.428	.21–.53	.436	.26–.43	.104	.01–.18	.308
WSRFNet	FGADR	.011	.01–.01	.158	.12–.17	.086	.05–.09	.020	.01–.03	.069
WSRFNet	Retinal-L.	.037	.03–.04	.085	.05–.10	.107	.07–.12	.062	.03–.09	.073
WSRFNet	DiaRetDB1	.050	.02–.07	.215	.15–.25	.178	.13–.19	.158	.08–.20	.150
CLC-Net	IDRiD	.358	.29–.37	.614	.47–.64	.830	.71–.86	.570	.25–.67	.593
CLC-Net	DDR	.152	.10–.18	.466	.34–.53	.612	.51–.64	.428	.25–.51	.414
CLC-Net	TJDR std	.352	.19–.42	.666	.55–.73	.667	.49–.73	.296	.09–.45	.495
CLC-Net	TJDR UWF	.110	.04–.21	.137	.02–.14	.159	.06–.28	.095	.02–.29	.125
CLC-Net	MAPLES	.287	.17–.31	.475	.26–.56	.611	.33–.61	.030	.00–.11	.351
CLC-Net	FGADR	.011	.01–.01	.304	.24–.32	.305	.22–.34	.089	.05–.11	.177
CLC-Net	Retinal-L.	.048	.04–.05	.164	.08–.16	.222	.12–.22	.173	.11–.20	.152
CLC-Net	DiaRetDB1	.040	.02–.06	.318	.16–.38	.260	.16–.23	.242	.05–.31	.215
TC-Net	IDRiD	.225	.20–.25	.593	.47–.67	.773	.67–.82	.636	.38–.76	.557
TC-Net	DDR	.112	.08–.14	.476	.40–.54	.607	.55–.65	.250	.12–.39	.361
TC-Net	TJDR std	.239	.12–.34	.598	.50–.67	.670	.52–.75	.355	.09–.52	.466

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Model	Test set	MA	95% CI	HE	95% CI	EX	95% CI	SE	95% CI	mAUPR
TC-Net	TJDR UWF	.100	.04–.22	.383	.20–.53	.234	.14–.35	.056	.02–.17	.193
TC-Net	MAPLES	.241	.17–.30	.469	.24–.61	.722	.48–.81	.027	.00–.09	.365
TC-Net	FGADR	.018	.02–.02	.334	.30–.37	.349	.29–.40	.156	.12–.20	.214
TC-Net	Retinal-L.	.056	.04–.06	.206	.16–.24	.314	.22–.35	.145	.10–.19	.180
TC-Net	DiaRetDB1	.023	.01–.04	.475	.37–.55	.509	.36–.53	.188	.05–.27	.299
SAMed	IDRiD	.373	.33–.42	.653	.52–.73	.845	.76–.89	.696	.49–.80	.642
SAMed	DDR	.173	.13–.22	.574	.52–.64	.669	.62–.72	.460	.30–.60	.469
SAMed	TJDR std	.428	.29–.51	.743	.65–.81	.707	.57–.79	.570	.26–.73	.612
SAMed	TJDR UWF	.130	.05–.28	.450	.23–.60	.208	.12–.32	.043	.01–.19	.208
SAMed	MAPLES	.365	.28–.44	.515	.30–.65	.732	.51–.82	.227	.07–.36	.460
SAMed	FGADR	.027	.02–.03	.412	.37–.45	.442	.39–.49	.279	.23–.34	.290
SAMed	Retinal-L.	.109	.09–.12	.316	.26–.37	.384	.30–.44	.304	.25–.35	.278
SAMed	DiaRetDB1	.047	.02–.07	.480	.38–.56	.468	.37–.52	.295	.12–.43	.322
SAM-Adapter	IDRiD	.411	.34–.42	.644	.51–.74	.804	.71–.85	.649	.42–.73	.627
SAM-Adapter	DDR	.147	.10–.18	.538	.43–.62	.592	.51–.63	.454	.30–.56	.433
SAM-Adapter	TJDR std	.358	.22–.43	.751	.67–.82	.625	.51–.67	.432	.19–.64	.542
SAM-Adapter	TJDR UWF	.111	.04–.17	.387	.20–.56	.193	.09–.31	.104	.02–.23	.199
SAM-Adapter	MAPLES	.277	.18–.31	.454	.25–.57	.570	.30–.61	.187	.08–.30	.372
SAM-Adapter	FGADR	.019	.02–.02	.384	.32–.41	.358	.29–.41	.219	.17–.26	.245
SAM-Adapter	Retinal-L.	.054	.04–.05	.217	.13–.19	.273	.15–.31	.228	.18–.25	.193
SAM-Adapter	DiaRetDB1	.071	.04–.10	.461	.28–.50	.412	.19–.33	.221	.05–.30	.291
SAM2-UNet	IDRiD	.351	.29–.36	.524	.43–.56	.782	.68–.81	.572	.23–.67	.557
SAM2-UNet	DDR	.153	.10–.18	.493	.36–.55	.518	.41–.56	.278	.11–.29	.360
SAM2-UNet	TJDR std	.325	.18–.37	.647	.55–.71	.640	.51–.69	.416	.11–.49	.507
SAM2-UNet	TJDR UWF	.099	.04–.18	.382	.20–.52	.219	.10–.30	.118	.03–.22	.204
SAM2-UNet	MAPLES	.331	.22–.38	.442	.23–.55	.715	.46–.76	.104	.01–.21	.398
SAM2-UNet	FGADR	.017	.01–.02	.356	.28–.37	.374	.29–.39	.241	.15–.24	.247
SAM2-UNet	Retinal-L.	.084	.06–.08	.215	.13–.20	.322	.21–.29	.301	.21–.31	.231
SAM2-UNet	DiaRetDB1	.040	.02–.06	.331	.23–.35	.409	.27–.35	.286	.08–.44	.267
MedSAM (full FT)	IDRiD	.331	.29–.37	.676	.55–.76	.806	.72–.86	.522	.00–.20	.584
MedSAM (full FT)	DDR	.136	.10–.19	.529	.42–.64	.626	.57–.68	.361	.07–.23	.413
MedSAM (full FT)	TJDR std	.347	.24–.42	.740	.65–.81	.686	.57–.77	.440	.00–.14	.553
MedSAM (full FT)	TJDR UWF	.129	.05–.24	.577	.34–.70	.225	.14–.33	.041	.00–.00	.243
MedSAM (full FT)	MAPLES	.369	.29–.44	.515	.29–.65	.687	.43–.81	.097	.00–.00	.417
MedSAM (full FT)	FGADR	.018	.02–.02	.419	.38–.46	.424	.37–.48	.209	.03–.07	.268
MedSAM (full FT)	Retinal-L.	.097	.08–.10	.359	.29–.39	.421	.33–.48	.174	.01–.05	.262
MedSAM (full FT)	DiaRetDB1	.045	.02–.07	.462	.36–.55	.487	.39–.55	.141	.00–.16	.284
RETFound + UPerNet	IDRiD	.446	.41–.48	.681	.57–.74	.845	.76–.88	.603	.33–.75	.644
RETFound + UPerNet	DDR	.189	.14–.24	.491	.39–.60	.640	.56–.71	.546	.38–.66	.466
RETFound + UPerNet	TJDR std	.387	.28–.46	.733	.65–.79	.724	.56–.81	.513	.17–.70	.589
RETFound + UPerNet	TJDR UWF	.073	.02–.17	.064	.02–.18	.278	.18–.39	.030	.01–.09	.111
RETFound + UPerNet	MAPLES	.420	.35–.47	.531	.30–.67	.722	.49–.81	.166	.03–.31	.460
RETFound + UPerNet	FGADR	.023	.02–.03	.419	.37–.45	.406	.34–.45	.228	.17–.29	.269
RETFound + UPerNet	Retinal-L.	.107	.08–.10	.321	.22–.35	.419	.30–.43	.243	.17–.29	.273
RETFound + UPerNet	DiaRetDB1	.043	.02–.06	.430	.30–.52	.477	.33–.48	.306	.12–.42	.314

TJDR UWF, ultra-widefield images predicted at native resolution with sliding windows. Retinal-L., Retinal-Lesions without its grey-level 127 regions. DiaRetDB1, pixel AUPR against the ≥ 0.75 expert consensus (the official image-level protocol is Supplementary Table 13).

Supplementary Table 5 Secondary metrics of every main-table model on the internal tests (mean over the four classes and over seeds). 11-thr., the 11-threshold AUPR of the M2MRF evaluation code on the same maps; Dice and IoU at the threshold chosen on the validation split; lesion-level precision and recall by lesion size (S, M, L: small, medium, large; Methods) at the same threshold; image-level AUC of the maximum probability inside the field of view.

Model	Seeds	mAUPR	11-thr.	Dice	IoU	Lesion prec.	Recall S	Recall M	Recall L	Image AUC
Trained on IDRiD (internal test; mean over seeds)										
U-Net	3	.644 ± .015	.638 ± .013	.622 ± .020	.465 ± .021	.489 ± .023	.654 ± .047	.851 ± .032	.800 ± .040	.797 ± .041
DeepLabv3	3	.537 ± .006	.536 ± .004	.515 ± .006	.366 ± .005	.551 ± .019	.270 ± .049	.533 ± .033	.776 ± .026	–
HRNet-W48	3	.700 ± .004	.691 ± .003	.627 ± .004	.468 ± .004	.485 ± .010	.584 ± .040	.879 ± .013	.862 ± .018	–
Swin-Unet	3	.688 ± .003	.686 ± .002	.640 ± .002	.481 ± .002	.419 ± .007	.671 ± .050	.849 ± .008	.827 ± .014	–
FCT	3	.298 ± .084	.359 ± .071	.330 ± .066	.217 ± .048	.267 ± .047	.205 ± .032	.523 ± .029	.726 ± .027	–
nnU-Net	3	.665 ± .008	.675 ± .006	.656 ± .009	.498 ± .010	.552 ± .016	.649 ± .044	.828 ± .003	.739 ± .004	.786 ± .016
nnU-Net ResEnc-M	1	.631	.631	.638	.479	.526	.600	.852	.695	–
M2MRF	3	.657 ± .010	.650 ± .004	.648 ± .004	.490 ± .004	.531 ± .008	.615 ± .015	.824 ± .009	.736 ± .018	–
HACDR-Net	3	.538 ± .031	.559 ± .013	.520 ± .025	.367 ± .022	.495 ± .040	.411 ± .019	.749 ± .046	.764 ± .014	.698 ± .163
HRDecoder	1	.647	.637	.636	.476	.605	.456	.792	.785	.885
PMCNet	1	.652	.648	.616	.458	.529	.592	.825	.804	.786
WSRFNet	1	.582	.605	.641	.482	.538	.611	.808	.701	.852
CLC-Net	1	.661	.665	.632	.475	.495	.565	.832	.864	.868
TC-Net	1	.653	.643	.607	.449	.536	.443	.804	.815	.874
SAMed	3	.706 ± .015	.703 ± .012	.658 ± .007	.502 ± .008	.519 ± .007	.553 ± .055	.798 ± .033	.838 ± .030	.907 ± .033
SAM-Adapter	1	.706	.704	.663	.508	.527	.624	.851	.809	.832
SAM2-UNet	1	.664	.683	.653	.495	.440	.846	.925	.826	.835
MedSAM (full FT)	1	.642	.654	.609	.451	.507	.392	.759	.791	.791
RETFound + UPerNet	3	.699 ± .009	.692 ± .008	.650 ± .008	.492 ± .009	.469 ± .014	.683 ± .015	.873 ± .014	.856 ± .028	.821 ± .040
Trained on DDR (internal test; mean over seeds)										
U-Net	3	.371 ± .009	.421 ± .008	.405 ± .017	.260 ± .014	.435 ± .042	.247 ± .016	.482 ± .012	.646 ± .009	.786 ± .008
DeepLabv3	3	.377 ± .005	.382 ± .003	.380 ± .006	.246 ± .004	.497 ± .015	.062 ± .005	.287 ± .020	.655 ± .010	–
HRNet-W48	3	.447 ± .009	.461 ± .009	.448 ± .004	.297 ± .004	.513 ± .006	.217 ± .007	.495 ± .006	.680 ± .012	–
Swin-Unet	3	.431 ± .001	.447 ± .003	.440 ± .003	.289 ± .003	.424 ± .008	.312 ± .012	.543 ± .013	.674 ± .002	–
FCT	1	.176	.344	.197	.125	.224	.096	.351	.608	–
nnU-Net	3	.431 ± .003	.467 ± .001	.458 ± .010	.304 ± .008	.515 ± .008	.258 ± .006	.470 ± .006	.622 ± .007	.796 ± .004
nnU-Net ResEnc-M	1	.428	.448	.463	.310	.491	.294	.490	.664	–
M2MRF	3	.369 ± .011	.406 ± .008	.407 ± .012	.262 ± .009	.454 ± .007	.276 ± .003	.556 ± .001	.696 ± .023	–
HACDR-Net	3	.252 ± .017	.289 ± .022	.289 ± .022	.175 ± .014	.416 ± .019	.106 ± .022	.350 ± .028	.562 ± .016	.719 ± .012
HRDecoder	1	.296	.333	.373	.236	.459	.234	.496	.629	.714
PMCNet	1	.312	.342	.338	.206	.431	.200	.454	.629	.770
WSRFNet	1	.234	.264	.341	.211	.431	.284	.458	.531	.659
CLC-Net	1	.414	.433	.462	.309	.506	.266	.470	.639	.781
TC-Net	1	.361	.365	.352	.223	.443	.102	.376	.634	.785
SAMed	3	.460 ± .027	.472 ± .027	.454 ± .025	.301 ± .022	.497 ± .008	.242 ± .029	.474 ± .031	.674 ± .007	.842 ± .005
SAM-Adapter	1	.433	.465	.457	.304	.471	.257	.496	.690	.657
SAM2-UNet	1	.360	.445	.413	.265	.459	.298	.559	.698	.749
MedSAM (full FT)	1	.413	.433	.432	.283	.516	.212	.484	.675	.795
RETFound + UPerNet	3	.450 ± .018	.458 ± .016	.447 ± .006	.294 ± .005	.472 ± .006	.290 ± .006	.538 ± .007	.696 ± .009	.844 ± .002

±, standard deviation over seeds where more than one seed was run.

Supplementary Table 6 Registered and revised recipes of the three models whose per-image loss term emptied a rare-class channel (seed 0, internal tests). The registered recipe is the one frozen before training (per-image Dice or IoU with smoothing 1); the revised recipe pools the term over the batch (Methods, rare-class collapse). The main tables carry the revised runs; the registered runs stay in the repository under the `_registered` suffix.

Model	Source	Recipe	MA	HE	EX	SE	mAUPR
U-Net	IDRiD	registered	.376	.534	.859	.605	.594
U-Net	IDRiD	revised (main table)	.406	.622	.896	.707	.658
U-Net	DDR	registered	.005	.031	.418	.000	.114
U-Net	DDR	revised (main table)	.141	.410	.553	.410	.378
SAM2-UNet	IDRiD	registered	.439	.665	.870	.571	.636
SAM2-UNet	IDRiD	revised (main table)	.439	.670	.871	.678	.664
SAM2-UNet	DDR	registered	.023	.265	.528	.022	.210
SAM2-UNet	DDR	revised (main table)	.153	.493	.518	.278	.360
MedSAM	IDRiD	registered	.382	.731	.867	.357	.584
MedSAM	IDRiD	revised (main table)	.373	.665	.855	.674	.642
MedSAM	DDR	registered	.000	.487	.573	.001	.265
MedSAM	DDR	revised (main table)	.136	.529	.626	.361	.413

Supplementary Table 7 Every factor cell, per class (seed 0). One factor is varied at a time on the representative models at the DDR pixel budget (Methods); the variant column names the level, the training-set column the data the cell was trained on. Each cell is listed on every test set it was applied to; in-source tests of merged cells are the rows whose test set was part of the training set.

Model	Level	Trained on	Test set	MA	HE	EX	SE	mAUPR
Training source								
M2MRF	bstd	FGADR	IDRiD	.141	.320	.729	.557	.437
M2MRF	bstd	FGADR	DDR	.055	.219	.404	.107	.197
M2MRF	bstd	FGADR	TJDR std	.267	.642	.666	.223	.449
M2MRF	bstd	FGADR	TJDR UWF	.035	.480	.282	.176	.243
M2MRF	bstd	FGADR	MAPLES	.086	.336	.492	.090	.251
M2MRF	bstd	FGADR	FGADR	.066	.511	.622	.238	.359
M2MRF	bstd	FGADR	Retinal-L.	.031	.179	.177	.167	.139
M2MRF	bstd	FGADR	DiaRetDB1	.006	.270	.287	.142	.176
M2MRF	bstd	TJDR-std	IDRiD	.139	.186	.719	.317	.340
M2MRF	bstd	TJDR-std	DDR	.082	.221	.516	.104	.231
M2MRF	bstd	TJDR-std	TJDR std	.325	.746	.654	.490	.554
M2MRF	bstd	TJDR-std	TJDR UWF	.022	.270	.131	.157	.145
M2MRF	bstd	TJDR-std	MAPLES	.117	.340	.375	.022	.214
M2MRF	bstd	TJDR-std	FGADR	.014	.161	.270	.099	.136
M2MRF	bstd	TJDR-std	Retinal-L.	.030	.103	.231	.084	.112
M2MRF	bstd	TJDR-std	DiaRetDB1	.018	.201	.295	.134	.162
RETFound + UPerNet	bstd	FGADR	IDRiD	.243	.563	.775	.582	.541
RETFound + UPerNet	bstd	FGADR	DDR	.019	.392	.470	.284	.291
RETFound + UPerNet	bstd	FGADR	TJDR std	.140	.752	.595	.357	.461
RETFound + UPerNet	bstd	FGADR	TJDR UWF	.073	.615	.336	.113	.284
RETFound + UPerNet	bstd	FGADR	MAPLES	.146	.492	.722	.118	.370
RETFound + UPerNet	bstd	FGADR	FGADR	.118	.651	.691	.370	.457
RETFound + UPerNet	bstd	FGADR	Retinal-L.	.115	.385	.467	.325	.323
RETFound + UPerNet	bstd	FGADR	DiaRetDB1	.019	.523	.590	.178	.327
RETFound + UPerNet	bstd	TJDR-std	IDRiD	.315	.506	.815	.275	.478
RETFound + UPerNet	bstd	TJDR-std	DDR	.136	.347	.614	.236	.333
RETFound + UPerNet	bstd	TJDR-std	TJDR std	.432	.823	.828	.741	.706
RETFound + UPerNet	bstd	TJDR-std	TJDR UWF	.053	.523	.150	.262	.247
RETFound + UPerNet	bstd	TJDR-std	MAPLES	.310	.330	.605	.131	.344
RETFound + UPerNet	bstd	TJDR-std	FGADR	.023	.327	.368	.185	.226
RETFound + UPerNet	bstd	TJDR-std	Retinal-L.	.085	.226	.336	.125	.193
RETFound + UPerNet	bstd	TJDR-std	DiaRetDB1	.027	.317	.491	.211	.262
SAM2-UNet	bstd	FGADR	IDRiD	.138	.560	.859	.099	.414
SAM2-UNet	bstd	FGADR	DDR	.062	.385	.537	.092	.269
SAM2-UNet	bstd	FGADR	TJDR std	.320	.752	.768	.001	.461
SAM2-UNet	bstd	FGADR	TJDR UWF	.060	.593	.270	.000	.231
SAM2-UNet	bstd	FGADR	MAPLES	.150	.477	.744	.000	.343
SAM2-UNet	bstd	FGADR	FGADR	.122	.630	.675	.126	.388
SAM2-UNet	bstd	FGADR	Retinal-L.	.043	.339	.411	.048	.210
SAM2-UNet	bstd	FGADR	DiaRetDB1	.012	.468	.516	.082	.270
SAM2-UNet	bstd	TJDR-std	IDRiD	.025	.439	.793	.026	.321
SAM2-UNet	bstd	TJDR-std	DDR	.016	.325	.542	.013	.224
SAM2-UNet	bstd	TJDR-std	TJDR std	.056	.753	.722	.102	.408
SAM2-UNet	bstd	TJDR-std	TJDR UWF	.008	.380	.095	.010	.123
SAM2-UNet	bstd	TJDR-std	MAPLES	.007	.281	.480	.000	.192
SAM2-UNet	bstd	TJDR-std	FGADR	.007	.294	.308	.005	.154
SAM2-UNet	bstd	TJDR-std	Retinal-L.	.010	.137	.201	.006	.088
SAM2-UNet	bstd	TJDR-std	DiaRetDB1	.000	.233	.230	.006	.117

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Model	Level	Trained on	Test set	MA	HE	EX	SE	mAUPR
U-Net	bstd	FGADR	IDRiD	.092	.294	.793	.276	.364
U-Net	bstd	FGADR	DDR	.032	.259	.467	.199	.239
U-Net	bstd	FGADR	TJDR std	.135	.718	.603	.339	.449
U-Net	bstd	FGADR	TJDR UWF	.028	.482	.208	.273	.248
U-Net	bstd	FGADR	MAPLES	.088	.337	.566	.064	.264
U-Net	bstd	FGADR	FGADR	.084	.568	.634	.325	.403
U-Net	bstd	FGADR	Retinal-L.	.084	.224	.368	.155	.208
U-Net	bstd	FGADR	DiaRetDB1	.005	.261	.563	.188	.254
U-Net	bstd	TJDR-std	IDRiD	.108	.324	.641	.435	.377
U-Net	bstd	TJDR-std	DDR	.071	.264	.531	.161	.257
U-Net	bstd	TJDR-std	TJDR std	.326	.775	.642	.455	.550
U-Net	bstd	TJDR-std	TJDR UWF	.015	.203	.077	.182	.119
U-Net	bstd	TJDR-std	MAPLES	.080	.327	.447	.033	.222
U-Net	bstd	TJDR-std	FGADR	.015	.237	.266	.076	.148
U-Net	bstd	TJDR-std	Retinal-L.	.028	.107	.193	.098	.107
U-Net	bstd	TJDR-std	DiaRetDB1	.008	.169	.217	.144	.134
nnU-Net	bstd	FGADR	IDRiD	.121	.596	.826	.717	.565
nnU-Net	bstd	FGADR	DDR	.063	.378	.532	.306	.320
nnU-Net	bstd	FGADR	TJDR std	.147	.737	.696	.520	.525
nnU-Net	bstd	FGADR	TJDR UWF	.021	.277	.222	.391	.228
nnU-Net	bstd	FGADR	MAPLES	.197	.517	.719	.138	.393
nnU-Net	bstd	FGADR	FGADR	.136	.631	.659	.372	.449
nnU-Net	bstd	FGADR	Retinal-L.	.039	.390	.439	.247	.279
nnU-Net	bstd	FGADR	DiaRetDB1	.028	.474	.559	.226	.322
nnU-Net	bstd	TJDR-std	IDRiD	.146	.459	.664	.474	.436
nnU-Net	bstd	TJDR-std	DDR	.079	.310	.550	.170	.277
nnU-Net	bstd	TJDR-std	TJDR std	.305	.785	.787	.668	.636
nnU-Net	bstd	TJDR-std	TJDR UWF	.013	.311	.124	.268	.179
nnU-Net	bstd	TJDR-std	MAPLES	.162	.462	.457	.033	.278
nnU-Net	bstd	TJDR-std	FGADR	.019	.322	.299	.157	.199
nnU-Net	bstd	TJDR-std	Retinal-L.	.031	.162	.224	.125	.136
nnU-Net	bstd	TJDR-std	DiaRetDB1	.027	.395	.338	.157	.229
Annotation granularity								
M2MRF	DDR	DDR	IDRiD	.392	.555	.776	.563	.571
M2MRF	DDR	DDR	TJDR std	.299	.696	.618	.412	.506
M2MRF	DDR	DDR	DiaRetDB1	.049	.218	.177	.247	.173
U-Net	DDR	DDR	IDRiD	.364	.541	.800	.483	.547
U-Net	DDR	DDR	TJDR std	.296	.676	.614	.392	.494
U-Net	DDR	DDR	DiaRetDB1	.039	.314	.268	.253	.219
M2MRF	FGADR	FGADR	IDRiD	.141	.320	.729	.557	.437
M2MRF	FGADR	FGADR	TJDR std	.267	.642	.666	.223	.449
M2MRF	FGADR	FGADR	DiaRetDB1	.006	.270	.287	.142	.176
U-Net	FGADR	FGADR	IDRiD	.092	.294	.793	.276	.364
U-Net	FGADR	FGADR	TJDR std	.135	.718	.603	.339	.449
U-Net	FGADR	FGADR	DiaRetDB1	.005	.261	.563	.188	.254
M2MRF	DDR + FGADR (383)	DDR + FGADR (383)	IDRiD	.386	.446	.799	.594	.556
M2MRF	DDR + FGADR (383)	DDR + FGADR (383)	DDR	.141	.365	.508	.328	.336
M2MRF	DDR + FGADR (383)	DDR + FGADR (383)	TJDR std	.285	.669	.661	.321	.484
M2MRF	DDR + FGADR (383)	DDR + FGADR (383)	TJDR UWF	.110	.461	.172	.180	.231
M2MRF	DDR + FGADR (383)	DDR + FGADR (383)	MAPLES	.252	.378	.482	.134	.311
M2MRF	DDR + FGADR (383)	DDR + FGADR (383)	Retinal-L.	.044	.158	.177	.176	.139
M2MRF	DDR + FGADR (383)	DDR + FGADR (383)	DiaRetDB1	.043	.197	.222	.182	.161
M2MRF	DDR + FGADR	DDR + FGADR	IDRiD	.403	.466	.782	.565	.554

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Model	Level	Trained on	Test set	MA	HE	EX	SE	mAUPR
M2MRF	DDR + FGADR	DDR + FGADR	DDR	.139	.366	.492	.282	.320
M2MRF	DDR + FGADR	DDR + FGADR	TJDR std	.312	.680	.659	.337	.497
M2MRF	DDR + FGADR	DDR + FGADR	TJDR UWF	.109	.474	.186	.264	.258
M2MRF	DDR + FGADR	DDR + FGADR	MAPLES	.256	.391	.431	.119	.299
M2MRF	DDR + FGADR	DDR + FGADR	Retinal-L.	.045	.168	.173	.160	.137
M2MRF	DDR + FGADR	DDR + FGADR	DiaRetDB1	.041	.208	.176	.151	.144
M2MRF	DDR + Retinal-Lesions	DDR + Retinal-Lesions	IDRiD	.369	.399	.784	.508	.515
M2MRF	DDR + Retinal-Lesions	DDR + Retinal-Lesions	DDR	.108	.296	.449	.291	.286
M2MRF	DDR + Retinal-Lesions	DDR + Retinal-Lesions	TJDR std	.145	.480	.378	.299	.325
M2MRF	DDR + Retinal-Lesions	DDR + Retinal-Lesions	TJDR UWF	.086	.419	.223	.080	.202
M2MRF	DDR + Retinal-Lesions	DDR + Retinal-Lesions	MAPLES	.131	.270	.408	.113	.231
M2MRF	DDR + Retinal-Lesions	DDR + Retinal-Lesions	FGADR	.010	.157	.186	.149	.126
M2MRF	DDR + Retinal-Lesions	DDR + Retinal-Lesions	DiaRetDB1	.048	.199	.308	.185	.185
M2MRF	Retinal-Lesions	Retinal-Lesions	IDRiD	.063	.133	.186	.285	.167
M2MRF	Retinal-Lesions	Retinal-Lesions	DDR	.024	.097	.130	.166	.104
M2MRF	Retinal-Lesions	Retinal-Lesions	TJDR std	.039	.112	.160	.287	.150
M2MRF	Retinal-Lesions	Retinal-Lesions	TJDR UWF	.015	.121	.053	.120	.077
M2MRF	Retinal-Lesions	Retinal-Lesions	MAPLES	.110	.101	.342	.074	.157
M2MRF	Retinal-Lesions	Retinal-Lesions	FGADR	.018	.121	.131	.129	.100
M2MRF	Retinal-Lesions	Retinal-Lesions	Retinal-L.	.352	.508	.650	.366	.469
M2MRF	Retinal-Lesions	Retinal-Lesions	DiaRetDB1	.022	.328	.774	.355	.370
U-Net	DDR + FGADR (383)	DDR + FGADR (383)	IDRiD	.375	.517	.828	.594	.578
U-Net	DDR + FGADR (383)	DDR + FGADR (383)	DDR	.137	.414	.548	.376	.369
U-Net	DDR + FGADR (383)	DDR + FGADR (383)	TJDR std	.299	.728	.556	.177	.440
U-Net	DDR + FGADR (383)	DDR + FGADR (383)	TJDR UWF	.101	.473	.151	.242	.242
U-Net	DDR + FGADR (383)	DDR + FGADR (383)	MAPLES	.305	.472	.502	.057	.334
U-Net	DDR + FGADR (383)	DDR + FGADR (383)	Retinal-L.	.068	.244	.258	.177	.187
U-Net	DDR + FGADR (383)	DDR + FGADR (383)	DiaRetDB1	.045	.329	.352	.161	.221
U-Net	DDR + FGADR	DDR + FGADR	IDRiD	.365	.551	.858	.613	.597
U-Net	DDR + FGADR	DDR + FGADR	DDR	.117	.430	.545	.402	.373
U-Net	DDR + FGADR	DDR + FGADR	TJDR std	.325	.762	.622	.342	.513
U-Net	DDR + FGADR	DDR + FGADR	TJDR UWF	.089	.466	.210	.294	.265
U-Net	DDR + FGADR	DDR + FGADR	MAPLES	.257	.468	.575	.133	.358
U-Net	DDR + FGADR	DDR + FGADR	Retinal-L.	.042	.219	.236	.186	.171
U-Net	DDR + FGADR	DDR + FGADR	DiaRetDB1	.034	.320	.302	.173	.207
U-Net	DDR + Retinal-Lesions	DDR + Retinal-Lesions	IDRiD	.279	.502	.786	.521	.522
U-Net	DDR + Retinal-Lesions	DDR + Retinal-Lesions	DDR	.090	.366	.518	.389	.341
U-Net	DDR + Retinal-Lesions	DDR + Retinal-Lesions	TJDR std	.237	.643	.490	.440	.452
U-Net	DDR + Retinal-Lesions	DDR + Retinal-Lesions	TJDR UWF	.057	.480	.231	.210	.245
U-Net	DDR + Retinal-Lesions	DDR + Retinal-Lesions	MAPLES	.269	.465	.474	.147	.339
U-Net	DDR + Retinal-Lesions	DDR + Retinal-Lesions	FGADR	.013	.202	.271	.133	.155
U-Net	DDR + Retinal-Lesions	DDR + Retinal-Lesions	DiaRetDB1	.052	.278	.386	.278	.248
U-Net	Retinal-Lesions	Retinal-Lesions	IDRiD	.091	.230	.353	.317	.248
U-Net	Retinal-Lesions	Retinal-Lesions	DDR	.039	.146	.239	.234	.164
U-Net	Retinal-Lesions	Retinal-Lesions	TJDR std	.087	.378	.238	.419	.281
U-Net	Retinal-Lesions	Retinal-Lesions	TJDR UWF	.026	.208	.091	.163	.122
U-Net	Retinal-Lesions	Retinal-Lesions	MAPLES	.168	.284	.470	.102	.256
U-Net	Retinal-Lesions	Retinal-Lesions	FGADR	.017	.136	.148	.064	.091
U-Net	Retinal-Lesions	Retinal-Lesions	Retinal-L.	.353	.553	.721	.390	.504
U-Net	Retinal-Lesions	Retinal-Lesions	DiaRetDB1	.045	.494	.874	.483	.474
Input resolution								
M2MRF	res:1440	DDR	DDR	.148	.402	.547	.422	.380
RETFound + UPerNet	res:1440	DDR	DDR	.189	.491	.640	.546	.466

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Model	Level	Trained on	Test set	MA	HE	EX	SE	mAUPR
HRDecoder	res:2880 (author canvas, equivalent epochs)	IDRiD	IDRiD	.423	.606	.864	.615	.627
M2MRF	res:512	DDR	DDR	.114	.411	.531	.378	.358
M2MRF	res:native patches	DDR	DDR	.140	.360	.510	.237	.312
RETFound + UPerNet	res:512	DDR	DDR	.092	.346	.628	.535	.400
RETFound + UPerNet	res:native patches	DDR	DDR	.196	.460	.628	.513	.449
Training amount								
M2MRF	size:DDR 100%	DDR	DDR	.148	.402	.547	.422	.380
M2MRF	size:DDR 100%	DDR	TJDR std	.299	.696	.618	.412	.506
M2MRF	size:DDR 100%	DDR	FGADR	.019	.310	.340	.181	.212
U-Net	size:DDR 100%	DDR	DDR	.141	.410	.553	.410	.378
U-Net	size:DDR 100%	DDR	TJDR std	.296	.676	.614	.392	.494
U-Net	size:DDR 100%	DDR	FGADR	.013	.214	.292	.162	.170
M2MRF	size:DDR 10%	DDR	IDRiD	.264	.481	.735	.413	.473
M2MRF	size:DDR 10%	DDR	DDR	.107	.383	.516	.222	.307
M2MRF	size:DDR 10%	DDR	TJDR std	.211	.625	.579	.225	.410
M2MRF	size:DDR 10%	DDR	TJDR UWF	.102	.221	.167	.032	.131
M2MRF	size:DDR 10%	DDR	MAPLES	.188	.345	.446	.067	.262
M2MRF	size:DDR 10%	DDR	FGADR	.011	.242	.187	.148	.147
M2MRF	size:DDR 10%	DDR	Retinal-L.	.036	.139	.164	.138	.119
M2MRF	size:DDR 10%	DDR	DiaRetDB1	.030	.190	.186	.205	.153
M2MRF	size:DDR 50%	DDR	IDRiD	.364	.500	.768	.566	.549
M2MRF	size:DDR 50%	DDR	DDR	.140	.364	.532	.332	.342
M2MRF	size:DDR 50%	DDR	TJDR std	.265	.672	.603	.450	.498
M2MRF	size:DDR 50%	DDR	TJDR UWF	.103	.365	.234	.045	.187
M2MRF	size:DDR 50%	DDR	MAPLES	.249	.427	.478	.127	.320
M2MRF	size:DDR 50%	DDR	FGADR	.014	.245	.338	.184	.195
M2MRF	size:DDR 50%	DDR	Retinal-L.	.046	.166	.184	.177	.143
M2MRF	size:DDR 50%	DDR	DiaRetDB1	.048	.220	.193	.246	.177
M2MRF	size:fine merged	DDR	IDRiD	.446	.531	.824	.674	.619
M2MRF	size:fine merged	DDR	DDR	.141	.393	.534	.405	.368
M2MRF	size:fine merged	DDR	TJDR std	.333	.746	.640	.540	.565
M2MRF	size:fine merged	DDR	TJDR UWF	.120	.472	.197	.339	.282
M2MRF	size:fine merged	DDR	MAPLES	.285	.462	.500	.150	.349
M2MRF	size:fine merged	DDR	FGADR	.020	.347	.333	.194	.224
M2MRF	size:fine merged	DDR	Retinal-L.	.049	.191	.184	.187	.153
M2MRF	size:fine merged	DDR	DiaRetDB1	.049	.266	.191	.256	.190
U-Net	size:DDR 10%	DDR	IDRiD	.239	.474	.784	.470	.492
U-Net	size:DDR 10%	DDR	DDR	.106	.337	.508	.334	.321
U-Net	size:DDR 10%	DDR	TJDR std	.197	.525	.479	.308	.378
U-Net	size:DDR 10%	DDR	TJDR UWF	.077	.243	.205	.072	.149
U-Net	size:DDR 10%	DDR	MAPLES	.243	.303	.542	.092	.295
U-Net	size:DDR 10%	DDR	FGADR	.010	.132	.190	.119	.112
U-Net	size:DDR 10%	DDR	Retinal-L.	.037	.119	.173	.165	.123
U-Net	size:DDR 10%	DDR	DiaRetDB1	.030	.180	.247	.204	.165
U-Net	size:DDR 50%	DDR	IDRiD	.303	.552	.811	.493	.540
U-Net	size:DDR 50%	DDR	DDR	.132	.407	.570	.383	.373
U-Net	size:DDR 50%	DDR	TJDR std	.273	.715	.530	.374	.473
U-Net	size:DDR 50%	DDR	TJDR UWF	.123	.338	.261	.075	.199
U-Net	size:DDR 50%	DDR	MAPLES	.240	.476	.564	.076	.339
U-Net	size:DDR 50%	DDR	FGADR	.008	.230	.355	.155	.187
U-Net	size:DDR 50%	DDR	Retinal-L.	.045	.192	.214	.189	.160
U-Net	size:DDR 50%	DDR	DiaRetDB1	.035	.307	.262	.234	.210
U-Net	size:fine merged	DDR	IDRiD	.432	.563	.859	.574	.607

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continued

Model	Level	Trained on	Test set	MA	HE	EX	SE	mAUPR
U-Net	size:fine merged	DDR	DDR	.154	.393	.582	.337	.367
U-Net	size:fine merged	DDR	TJDR std	.367	.779	.643	.458	.562
U-Net	size:fine merged	DDR	TJDR UWF	.118	.489	.179	.288	.268
U-Net	size:fine merged	DDR	MAPLES	.295	.491	.596	.107	.372
U-Net	size:fine merged	DDR	FGADR	.014	.276	.322	.154	.192
U-Net	size:fine merged	DDR	Retinal-L.	.050	.229	.240	.191	.177
U-Net	size:fine merged	DDR	DiaRetDB1	.041	.338	.292	.202	.218
M2MRF	size:DDR 10%:fov	DDR	TJDR UWF	.087	.304	.076	.096	.141
M2MRF	size:DDR 50%:fov	DDR	TJDR UWF	.115	.432	.160	.163	.218
U-Net	size:DDR 10%:fov	DDR	TJDR UWF	.047	.238	.080	.121	.122
U-Net	size:DDR 50%:fov	DDR	TJDR UWF	.065	.412	.187	.221	.221
Loss								
U-Net	loss:CE + Dice	DDR	DDR	.141	.410	.553	.410	.378
U-Net	loss:CE	DDR	DDR	.126	.429	.602	.271	.357
U-Net	loss:Dice	DDR	DDR	.124	.268	.397	.160	.237
U-Net	loss:CE + per-image Dice	DDR	DDR	.006	.010	.460	.000	.119

Supplementary Table 8 Computational cost. Parameters and trainable parameters under the unified protocol, training time of one seed on each source (40 GB A100, one task per card), and, where the cost audit has run, floating-point operations for one 1,440-pixel canvas, peak training memory and inference time per image.

Model	Params (M)	Trainable (M)	GPU h IDRiD	GPU h DDR	GFLOPs	Peak mem. (GB)	Infer. (s/image)
U-Net	7.8	–	1.5	5.4	–	–	–
DeepLabv3	58.6	–	3.6	12.8	–	–	–
HRNet-W48	65.8	–	2.6	10.1	–	–	–
Swin-Unet	27.2	–	2.6	10.4	–	–	–
FCT	1.9	–	19.7	72.0	–	–	–
nnU-Net	59.2	–	7.2	7.2	–	–	–
nnU-Net ResEnc-M	105.9	–	12.8	11.4	–	–	–
M2MRF	70.3	–	3.0	11.6	–	–	–
HACDR-Net	24.6	–	5.9	27.8	–	–	–
HRDecoder	65.6	–	1.5	5.9	–	–	–
PMCNet	12.9	–	1.8	7.1	–	–	–
WSRFNet	9.7	–	2.7	10.5	–	–	–
CLC-Net	119.7	–	9.9	39.2	–	–	–
TC-Net	14.9	–	6.0	24.7	–	–	–
SAMed	91.6	–	2.1	7.4	–	–	–
SAM-Adapter	641.3	4.3	6.5	26.3	–	–	–
SAM2-UNet	216.5	4.4	4.1	16.2	–	–	–
MedSAM (full FT)	93.7	93.7	2.8	11.1	–	–	–
RETFound + UPerNet	352.6	352.6	6.1	24.2	–	–	–

nnU-Net follows its own schedule of 1,000 epochs rather than the pixel budget. Cells without a value were not measured; the cost audit (FLOPs, memory, inference) is pending.

Supplementary Table 9 Refined IDRiD (label revision): per-class AUPR of every IDRiD-trained model against the official and the refined labels through the same 1,024-pixel geometry. Test maps of the main-table checkpoints (seed 0), moved through the recovered crop and scored with the protocol evaluator against both label sets; no model is retrained or re-run.

Model	MA off.	MA ref.	HE off.	HE ref.	EX off.	EX ref.	SE off.	SE ref.	mAUPR off.	mAUPR ref.	Δ
CLC-Net	.423	.420	.686	.685	.890	.889	.641	.640	.660	.659	−0.001
DeepLabv3	.091	.091	.643	.643	.736	.736	.664	.664	.533	.533	+0.000
FCT	.079	.078	.165	.165	.687	.686	.115	.115	.261	.261	−0.000
HACDR-Net	.358	.356	.504	.504	.828	.827	.561	.560	.563	.561	−0.001
HRDecoder	.407	.405	.634	.633	.827	.826	.730	.730	.649	.649	−0.001
HRNet-W48	.442	.438	.685	.684	.885	.883	.783	.783	.699	.697	−0.002
M2MRF	.439	.436	.625	.625	.879	.878	.681	.680	.656	.655	−0.001
MedSAM (full FT)	.370	.368	.666	.666	.858	.858	.676	.676	.643	.642	−0.001
PMCNet	.411	.409	.610	.610	.883	.881	.698	.698	.651	.649	−0.001
RETFound + UPerNet	.483	.480	.719	.718	.896	.895	.736	.736	.709	.707	−0.001
SAM2-UNet	.423	.421	.683	.682	.875	.874	.681	.681	.666	.665	−0.001
SAM-Adapter	.442	.439	.745	.745	.887	.887	.756	.756	.708	.707	−0.001
SAMed	.445	.442	.737	.737	.893	.892	.791	.790	.717	.715	−0.001
Swin-Unet	.473	.471	.712	.712	.895	.894	.689	.689	.692	.691	−0.001
TC-Net	.353	.351	.651	.651	.855	.854	.755	.755	.654	.653	−0.001
U-Net	.386	.384	.623	.622	.895	.894	.710	.711	.653	.653	−0.001
WSRFNet	.388	.386	.602	.602	.856	.855	.506	.506	.588	.587	−0.001
nnU-Net	.482	.480	.673	.673	.878	.876	.645	.644	.669	.668	−0.001
nnU-Net ResEnc-M	.442	.440	.667	.666	.878	.877	.541	.540	.632	.631	−0.001

Label differences over the 27 test images: MA: 13 images changed, 28,847 → 28,886 labelled pixels, 1,083 → 1,083 components; HE: 13 images changed, 311,289 → 311,524 labelled pixels, 532 → 532 components; EX: 16 images changed, 320,363 → 320,160 labelled pixels, 3,895 → 3,882 components; SE: 6 images changed, 53,684 → 53,597 labelled pixels, 38 → 38 components. The crop geometry of every image (all 27: padded to a square and resized; NCC $\geq$ 0.9988) is runs/REFINED/RefinedIDRiD_geometry.json.

Supplementary Table 10 Encoder comparison: four ViT-L encoders under one UPerNet head, recipe and pixel budget (seed 0). Per-class AUPR on the internal test and on the three independent fine-grained sets. RETFound-MAE is the main-table RETFound run of the same source; the other three are the control cells. The pre-specified contrasts are RETFound-MAE minus MAE-IN1k and RETFound-DINOv2 minus DINOv2.

Encoder	Test set	MA	HE	EX	SE	mAUPR
Trained on IDRiD						
RETFound-DINOv2	IDRiD	.502	.759	.900	.683	.711
RETFound-DINOv2	TJDR std	.375	.626	.714	.520	.559
RETFound-DINOv2	TJDR UWF	.092	.484	.211	.104	.223
RETFound-DINOv2	MAPLES	.380	.495	.682	.109	.417
RETFound-MAE	IDRiD	.484	.718	.895	.734	.708
RETFound-MAE	TJDR std	.393	.582	.776	.070	.455
RETFound-MAE	TJDR UWF	.099	.494	.106	.311	.253
RETFound-MAE	MAPLES	.375	.503	.680	.092	.413
DINOv2 ViT-L	IDRiD	.515	.765	.905	.772	.739
DINOv2 ViT-L	TJDR std	.341	.658	.676	.427	.525
DINOv2 ViT-L	TJDR UWF	.102	.557	.239	.091	.247
DINOv2 ViT-L	MAPLES	.392	.490	.693	.194	.442
MAE-IN1k ViT-L	IDRiD	.493	.744	.900	.778	.729
MAE-IN1k ViT-L	TJDR std	.392	.547	.709	.507	.539
MAE-IN1k ViT-L	TJDR UWF	.099	.586	.244	.147	.269
MAE-IN1k ViT-L	MAPLES	.370	.469	.659	.203	.425
Trained on DDR						
RETFound-DINOv2	DDR	.184	.508	.657	.611	.490
RETFound-DINOv2	TJDR std	.354	.786	.716	.607	.616
RETFound-DINOv2	TJDR UWF	.113	.264	.187	.042	.151
RETFound-DINOv2	MAPLES	.422	.522	.731	.307	.495
RETFound-MAE	DDR	.189	.491	.640	.546	.466
RETFound-MAE	TJDR std	.387	.733	.724	.513	.589
RETFound-MAE	TJDR UWF	.073	.064	.278	.030	.111
RETFound-MAE	MAPLES	.420	.531	.722	.166	.460
DINOv2 ViT-L	DDR	.183	.572	.668	.618	.510
DINOv2 ViT-L	TJDR std	.390	.791	.698	.596	.619
DINOv2 ViT-L	TJDR UWF	.100	.133	.265	.050	.137
DINOv2 ViT-L	MAPLES	.425	.523	.728	.274	.488
MAE-IN1k ViT-L	DDR	.206	.567	.670	.634	.519
MAE-IN1k ViT-L	TJDR std	.422	.788	.717	.611	.634
MAE-IN1k ViT-L	TJDR UWF	.103	.100	.244	.061	.127
MAE-IN1k ViT-L	MAPLES	.406	.515	.726	.265	.478

Supplementary Table 11 Ensembles of the main-table probability maps and the reference model, mAUPR on every test set. Compositions fixed before any ensemble was scored (Methods): all 18 registered members, the five and three best by validation mAUPR, the best of each family, and each family alone. The reference model's two cells are trained on the three fine-grained sources merged; their ensemble is the pixel mean of their maps.

Composition	IDRiD	DDR	TJDR std	TJDR UWF	MAPLES	FGADR	Retinal-L.	DiaRetDB1
Members trained on IDRiD								
E-all	.764	.400	.594	.311	.455	.279[p]	.216[p]	.317
E-top5	.754	.410	.598	.292	.449	.290	.251	.316
E-top3	.742	.394	.559	.279	.437	.273	.252	.325
E-1perfam	.741	.381	.555	.284	.445	.256	.239	.318
E-fam:general	.727	.352	.513	.276	.410	.244	.194	.302
E-fam:auto	.693	.316	.485	.229	.406	.242	.177	.269
E-fam:fundus	.736	.371	.568	.292	.436	.250[p]	.214[p]	.300
E-fam:foundation	.757	.411	.609	.297	.442	.294	.250	.318
Members trained on DDR								
E-all	.677	.505	.637	.266	.496	.291[p]	.278[p]	.335
E-top5	.680	.511	.652	.242	.506	.298	.290	.335
E-top3	.672	.510	.655	.226	.497	.300	.291	.339
E-1perfam	.665	.494	.625	.228	.486	.281	.278	.321
E-fam:general	.650	.483	.606	.227	.470	.268	.266	.324
E-fam:auto	.641	.463	.572	.210	.473	.275	.227	.334
E-fam:fundus	.637	.444	.568	.246	.452	.251[p]	.218[p]	.305
E-fam:foundation	.676	.509	.657	.248	.494	.302	.294	.340
Reference model: trained on IDRiD + DDR + TJDR std merged (DDR budget and validation)								
SAMed (fine merged)	–	.477	–	–	–	–	–	–

[p], computed without HACDR-Net, whose maps for that set could not be regenerated (members named in the run record). In-source test sets of the reference model (IDRiD, DDR, TJDR std) are the official test splits of sources that entered its training.

Supplementary Table 12 Results by DR grade (seed 0). On the test sets that carry a DR grade per image, mAUPR and image-level AUC recomputed within grades 0–1 and within grades 2–4 from the stored per-image histograms, with the thresholds of the full test; ungradable images enter neither stratum.

Model	Trained on	mAUPR 0–1	mAUPR 2–4	Image AUC 0–1	Image AUC 2–4	Δ mAUPR
DDR: 20 images of grades 0–1, 203 of grades 2–4						
CLC-Net	DDR	.254	.384	.653	.752	+0.129
CLC-Net	IDRiD	.147	.295	.486	.667	+0.148
DeepLabv3	DDR	.145	.372	–	–	+0.227
DeepLabv3	IDRiD	.169	.258	.486	.720	+0.090
FCT	DDR	.001	.177	–	–	+0.176
FCT	IDRiD	.002	.078	.486	.597	+0.076
HACDR-Net	DDR	.210	.268	–	–	+0.058
HACDR-Net	IDRiD	.196	.163	.514	.651	−0.032
HRDecoder	DDR	.266	.273	.639	.692	+0.007
HRDecoder	IDRiD	.128	.276	.542	.658	+0.148
HRNet-W48	DDR	.260	.442	–	–	+0.182
HRNet-W48	IDRiD	.257	.336	.444	.772	+0.079
M2MRF	DDR	.209	.349	–	–	+0.140
M2MRF	IDRiD	.120	.218	.542	.628	+0.099
MedSAM (full FT)	DDR	.225	.361	.444	.782	+0.137
MedSAM (full FT)	IDRiD	.189	.315	.639	.757	+0.126
PMCNet	DDR	.213	.302	.611	.750	+0.089
PMCNet	IDRiD	.193	.262	.167	.701	+0.069
RETFound + UPerNet	DDR	.256	.462	.597	.829	+0.206
RETFound + UPerNet	IDRiD	.183	.355	.292	.797	+0.172
SAM2-UNet	DDR	.270	.315	.514	.741	+0.044
SAM2-UNet	IDRiD	.156	.283	.604	.615	+0.127
SAM-Adapter	DDR	.273	.416	.583	.639	+0.143
SAM-Adapter	IDRiD	.189	.292	.625	.590	+0.103
SAMed	DDR	.227	.466	.361	.832	+0.240
SAMed	IDRiD	.189	.381	.576	.778	+0.192
Swin-Unet	DDR	.251	.431	–	–	+0.179
Swin-Unet	IDRiD	.157	.329	.340	.744	+0.172
TC-Net	DDR	.185	.360	.417	.761	+0.176
TC-Net	IDRiD	.197	.313	.500	.772	+0.116
U-Net	DDR	.206	.341	.458	.776	+0.135
U-Net	IDRiD	.112	.265	.222	.715	+0.153
WSRFNet	DDR	.160	.217	.569	.637	+0.057
WSRFNet	IDRiD	.123	.236	.479	.609	+0.113
nnU-Net	DDR	.150	.410	–	–	+0.260
nnU-Net	IDRiD	.124	.274	.514	.702	+0.150
nnU-Net ResEnc-M	DDR	.151	.403	–	–	+0.252
nnU-Net ResEnc-M	IDRiD	.150	.249	.542	.750	+0.098
MAPLES: 56 images of grades 0–1, 4 of grades 2–4						
CLC-Net	DDR	.297	.308	.650	.667	+0.011
CLC-Net	IDRiD	.349	.445	.638	1.000	+0.096
DeepLabv3	DDR	.332	.366	.752	1.000	+0.034
DeepLabv3	IDRiD	.261	.425	.693	1.000	+0.164
FCT	DDR	.127	.186	.470	.333	+0.060
FCT	IDRiD	.107	.177	.521	1.000	+0.069
HACDR-Net	DDR	.290	.371	.613	1.000	+0.081
HACDR-Net	IDRiD	.265	.407	.635	1.000	+0.143
HRDecoder	DDR	.265	.284	.761	.833	+0.019
HRDecoder	IDRiD	.338	.436	.650	1.000	+0.098
HRNet-W48	DDR	.433	.459	.788	.667	+0.027
HRNet-W48	IDRiD	.379	.465	.638	1.000	+0.086
M2MRF	DDR	.292	.329	.690	1.000	+0.037
M2MRF	IDRiD	.280	.330	.638	1.000	+0.050
MedSAM (full FT)	DDR	.373	.425	.743	.667	+0.051
MedSAM (full FT)	IDRiD	.337	.478	.738	1.000	+0.142
PMCNet	DDR	.370	.434	.735	.667	+0.064
PMCNet	IDRiD	.325	.436	.679	1.000	+0.110
RETFound + UPerNet	DDR	.437	.494	.830	.667	+0.056
RETFound + UPerNet	IDRiD	.386	.509	.721	.667	+0.123
SAM2-UNet	DDR	.354	.403	.675	.833	+0.049
SAM2-UNet	IDRiD	.332	.391	.658	.833	+0.059
SAM-Adapter	DDR	.325	.358	.657	.833	+0.032
SAM-Adapter	IDRiD	.334	.350	.677	1.000	+0.016
SAMed	DDR	.454	.471	.807	1.000	+0.018
SAMed	IDRiD	.393	.495	.792	1.000	+0.101
Swin-Unet	DDR	.435	.548	.770	1.000	+0.113
Swin-Unet	IDRiD	.374	.456	.622	.667	+0.083
TC-Net	DDR	.324	.432	.735	.667	+0.108
TC-Net	IDRiD	.383	.522	.701	1.000	+0.138
U-Net	DDR	.262	.375	.699	1.000	+0.114
U-Net	IDRiD	.277	.374	.681	1.000	+0.097
WSRFNet	DDR	.257	.327	.669	.667	+0.071
WSRFNet	IDRiD	.264	.319	.630	.333	+0.054
nnU-Net	DDR	.343	.427	.818	1.000	+0.084

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continued

Model	Trained on	mAUPR 0–1	mAUPR 2–4	Image AUC 0–1	Image AUC 2–4	Δ mAUPR
nnU-Net	IDRiD	.338	.420	.700	1.000	+0.082
nnU-Net ResEnc-M	DDR	.375	.423	.797	1.000	+0.048
nnU-Net ResEnc-M	IDRiD	.301	.350	.696	.333	+0.050
FGADR: 94 images of grades 0–1, 459 of grades 2–4						
CLC-Net	DDR	.059	.166	.560	.747	+0.107
CLC-Net	IDRiD	.016	.172	.633	.698	+0.156
DeepLabv3	DDR	.056	.236	.762	.753	+0.179
DeepLabv3	IDRiD	.055	.222	.661	.762	+0.167
FCT	DDR	.003	.071	.571	.602	+0.068
FCT	IDRiD	.002	.052	.577	.634	+0.050
HACDR-Net	DDR	.012	.106	.443	.674	+0.095
HACDR-Net	IDRiD	.005	.038	.501	.573	+0.032
HRDecoder	DDR	.021	.188	.707	.727	+0.168
HRDecoder	IDRiD	.024	.199	.650	.715	+0.176
HRNet-W48	DDR	.046	.274	.775	.814	+0.229
HRNet-W48	IDRiD	.036	.256	.759	.802	+0.220
M2MRF	DDR	.037	.202	.714	.731	+0.165
M2MRF	IDRiD	.002	.084	.581	.599	+0.082
MedSAM (full FT)	DDR	.074	.231	.768	.803	+0.157
MedSAM (full FT)	IDRiD	.031	.270	.635	.759	+0.238
PMCNet	DDR	.036	.211	.610	.793	+0.175
PMCNet	IDRiD	.027	.216	.730	.752	+0.189
RETFound + UPerNet	DDR	.061	.268	.700	.814	+0.207
RETFound + UPerNet	IDRiD	.017	.240	.719	.773	+0.222
SAM2-UNet	DDR	.028	.222	.712	.754	+0.195
SAM2-UNet	IDRiD	.037	.235	.734	.759	+0.198
SAM-Adapter	DDR	.035	.242	.687	.669	+0.207
SAM-Adapter	IDRiD	.014	.221	.564	.620	+0.207
SAMed	DDR	.067	.295	.770	.832	+0.228
SAMed	IDRiD	.053	.289	.745	.800	+0.236
Swin-Unet	DDR	.043	.248	.700	.802	+0.204
Swin-Unet	IDRiD	.017	.204	.590	.726	+0.187
TC-Net	DDR	.039	.219	.735	.795	+0.180
TC-Net	IDRiD	.020	.221	.640	.782	+0.201
U-Net	DDR	.019	.170	.690	.761	+0.151
U-Net	IDRiD	.003	.144	.576	.700	+0.141
WSRFNet	DDR	.016	.062	.652	.639	+0.046
WSRFNet	IDRiD	.016	.187	.642	.651	+0.170
nnU-Net	DDR	.046	.249	.728	.773	+0.203
nnU-Net	IDRiD	.036	.222	.673	.784	+0.186
nnU-Net ResEnc-M	DDR	.056	.245	.794	.810	+0.190
nnU-Net ResEnc-M	IDRiD	.027	.189	.615	.773	+0.162
Retinal-L.: 150 images of grades 0–1, 328 of grades 2–4						
CLC-Net	DDR	.057	.125	.762	.807	+0.068
CLC-Net	IDRiD	.029	.168	.807	.768	+0.139
DeepLabv3	DDR	.097	.273	.730	.798	+0.176
DeepLabv3	IDRiD	.051	.286	.695	.746	+0.236
FCT	DDR	.003	.085	.612	.605	+0.081
FCT	IDRiD	.002	.035	.529	.570	+0.033
HACDR-Net	DDR	.038	.140	.670	.782	+0.103
HACDR-Net	IDRiD	.012	.086	.684	.662	+0.074
HRDecoder	DDR	.050	.116	.849	.793	+0.066
HRDecoder	IDRiD	.030	.130	.767	.760	+0.100
HRNet-W48	DDR	.112	.264	.792	.846	+0.152
HRNet-W48	IDRiD	.061	.217	.780	.814	+0.156
M2MRF	DDR	.068	.135	.872	.831	+0.067
M2MRF	IDRiD	.023	.115	.744	.727	+0.092
MedSAM (full FT)	DDR	.058	.225	.834	.839	+0.167
MedSAM (full FT)	IDRiD	.042	.240	.790	.786	+0.197
PMCNet	DDR	.055	.179	.808	.833	+0.125
PMCNet	IDRiD	.039	.185	.740	.754	+0.146
RETFound + UPerNet	DDR	.121	.248	.826	.856	+0.127
RETFound + UPerNet	IDRiD	.040	.221	.774	.833	+0.182
SAM2-UNet	DDR	.081	.194	.807	.777	+0.113
SAM2-UNet	IDRiD	.034	.168	.754	.730	+0.134
SAM-Adapter	DDR	.074	.169	.789	.730	+0.094
SAM-Adapter	IDRiD	.020	.170	.612	.694	+0.151
SAMed	DDR	.125	.281	.828	.864	+0.155
SAMed	IDRiD	.055	.268	.795	.806	+0.213
Swin-Unet	DDR	.100	.263	.846	.870	+0.163
Swin-Unet	IDRiD	.052	.247	.810	.822	+0.195
TC-Net	DDR	.034	.182	.752	.808	+0.148
TC-Net	IDRiD	.026	.193	.768	.805	+0.167
U-Net	DDR	.069	.123	.850	.840	+0.054
U-Net	IDRiD	.028	.126	.778	.761	+0.098
WSRFNet	DDR	.032	.070	.705	.708	+0.037
WSRFNet	IDRiD	.036	.124	.756	.695	+0.087
nnU-Net	DDR	.073	.165	.858	.837	+0.092
nnU-Net	IDRiD	.028	.161	.835	.797	+0.132

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Model	Trained on	mAUPR 0–1	mAUPR 2–4	Image AUC 0–1	Image AUC 2–4	Δ mAUPR
nnU-Net ResEnc-M	DDR	.089	.172	.781	.873	+0.082
nnU-Net ResEnc-M	IDRiD	.020	.126	.798	.821	+0.106

Image-level AUC within a stratum is undefined when every image of the stratum carries the class or none does; such cells read 0 or 1 in the record and should be read with the stratum size.

Supplementary Table 13 Human reference on DiaRetDB1: each expert against the consensus of the other three. The expert's own markings, with their confidence levels mapped to scores, are the probability map and the consensus of the other three experts (≥ 0.75) is the ground truth; thresholds come from the official training list. A map with four confidence levels gives a four-point precision–recall curve, so the pixel-level AUPR is a lower bound.

Reader	MA	HE	EX	SE	mAUPR	Image AUC	Dice
Expert 01	.008	.246	.414	.102	.193	.870	.349
Expert 02	.001	.166	.273	.162	.150	.931	.261
Expert 03	.015	.341	.516	.244	.279	.931	.448
Expert 04	.022	.271	.366	.220	.219	.913	.360
Models trained on IDRiD, against the four-expert consensus (median over 19 models; range in brackets)							
models	–	–	–	–	0.245 [0.10–0.32]	0.833 [0.71–0.92]	–
Models trained on DDR, against the four-expert consensus (median over 19 models; range in brackets)							
models	–	–	–	–	0.284 [0.09–0.32]	0.840 [0.74–0.94]	–

Supplementary Table 14 Annotation style of every source: connected components per image and their size. All images of each source, every id once; 8-connected components of the prepared masks with ignore regions removed; the relative diameter is the component's equivalent diameter divided by the long side of the field-of-view box, in percent (Methods, annotation style).

Source	Class	Images	Components	Per image	Per positive image	Positive images	Diam. p10 (%)	p50 (%)	p90 (%)	Area px (median)
IDRiD	MA	81	3,497	43.2	43.2	81	0.33	0.50	0.78	234
IDRiD	HE	81	1,900	23.5	23.8	80	0.70	1.32	3.29	1608
IDRiD	EX	81	11,642	143.7	143.7	81	0.26	0.48	1.20	217
IDRiD	SE	81	150	1.9	3.8	40	1.86	3.07	5.45	8653
DDR	MA	757	10,380	13.7	18.2	570	0.21	0.43	0.77	48
DDR	HE	757	12,538	16.6	20.9	601	0.27	0.87	2.36	209
DDR	EX	757	23,659	31.3	48.7	486	0.16	0.45	1.31	56
DDR	SE	757	1,293	1.7	5.4	239	0.24	1.91	3.83	855
TJDR_std	MA	257	742	2.9	7.5	99	0.41	0.80	1.16	210
TJDR_std	HE	257	1,192	4.6	8.1	148	0.94	1.78	3.85	1048
TJDR_std	EX	257	1,130	4.4	11.0	103	0.62	1.26	2.76	525
TJDR_std	SE	257	216	0.8	3.3	65	1.36	3.11	6.19	3178
TJDR_uwf	MA	304	765	2.5	10.2	75	0.21	0.35	0.59	148
TJDR_uwf	HE	304	2,012	6.6	11.8	170	0.35	0.84	2.14	841
TJDR_uwf	EX	304	2,222	7.3	10.1	219	0.15	0.40	1.19	188
TJDR_uwf	SE	304	287	0.9	2.3	127	0.28	1.53	2.64	2826
FGADR	MA	1,842	11,235	6.1	7.9	1424	0.46	0.78	3.05	79
FGADR	HE	1,842	37,958	20.6	26.1	1456	0.59	1.17	3.45	176
FGADR	EX	1,842	52,896	28.7	41.4	1279	0.37	0.76	1.77	75
FGADR	SE	1,842	2,087	1.1	3.3	627	1.09	3.21	6.40	1323
Retinal-Lesions	MA	1,593	14,873	9.3	11.0	1346	1.37	2.31	4.10	336
Retinal-Lesions	HE	1,593	13,305	8.4	12.5	1064	1.82	3.45	7.12	749
Retinal-Lesions	EX	1,593	3,416	2.1	4.2	807	1.87	4.16	12.49	1089
Retinal-Lesions	SE	1,593	1,267	0.8	2.7	471	2.94	5.08	8.81	1624
DiaRetDB1	MA	89	183	2.1	4.6	40	0.48	0.62	0.72	61
DiaRetDB1	HE	89	402	4.5	10.1	40	0.71	2.93	8.61	1363
DiaRetDB1	EX	89	137	1.5	3.6	38	0.71	4.06	16.01	2624
DiaRetDB1	SE	89	27	0.3	1.8	15	1.90	4.31	8.02	2945
MAPLES-DR	MA	198	5,412	27.3	28.3	191	0.63	0.84	1.29	74
MAPLES-DR	HE	198	946	4.8	7.8	122	1.06	1.89	3.19	446
MAPLES-DR	EX	198	3,080	15.6	29.6	104	0.65	0.99	2.07	122
MAPLES-DR	SE	198	199	1.0	3.3	61	1.29	2.48	4.29	620

Supplementary Table 15 Near-duplicate audit between training sources and test sets, and the images excluded. Vessel signatures (green-channel black top-hat at 192 × 192 inside the field of view) are correlated between every train or validation image of a source and every test image; a test image at or above the pair's threshold is left out of that source's independent tests (default 0.5; 0.33 within FGADR, whose images are near-duplicates of each other at scale).

Training source	Test set	Excluded images
TJDR_uwf	TJDR_uwf	2
FGADR	FGADR	106
FGADR	Retinal-Lesions	19
Retinal-Lesions	FGADR	27
Retinal-Lesions	Retinal-Lesions	1
IDRiD	every test set	0
DDR	every test set	0
TJDR_std	every test set	0

Audit: bench/scripts/leak_audit.py –min-vessel 0.3 (all seven prepared sources), 4923 images. Strongest pairs (vessel NCC): FGADR train:0224_1 ↔ FGADR 0271_1 (1.00); FGADR train:0322_1 ↔ FGADR 0741_1 (1.00); FGADR train:0526_1 ↔ FGADR 1597_1 (1.00); FGADR train:0878_1 ↔ FGADR 0988_1 (1.00); FGADR val:0553_1 ↔ FGADR 0995_1 (1.00); TJDR_uwf train:TJDR_train_092 ↔ TJDR_uwf TJDR_test_021 (1.00); TJDR_uwf val:TJDR_train_101 ↔ TJDR_uwf TJDR_test_023 (1.00); Retinal-Lesions train:4428_left ↔ FGADR 0706_3 (0.99); Retinal-Lesions val:31746_right ↔ FGADR 0466_3 (0.98); Retinal-Lesions train:24235_right ↔ FGADR 0785_3 (0.98); Retinal-Lesions train:658_right ↔ FGADR 0236_3 (0.98); Retinal-Lesions train:15684_right ↔ FGADR 0627_3 (0.98). The full lists are bench/data/splits/leak_exclusions.json.

Supplementary Table 16 Released checkpoints on every test set, per class, under this protocol. Each checkpoint is run through its authors' own test pipeline (resize, normalisation, whole image or sliding window as configured) and the maps are scored with the protocol evaluator; a source the checkpoint was trained on is its internal test.

Checkpoint	Trained on	Test set	MA	HE	EX	SE	mAUPR
M2MRF	IDRiD	IDRiD	.408	.654	.832	.664	.639
M2MRF	IDRiD	DDR	.122	.289	.506	.061	.244
M2MRF	IDRiD	TJDR std	.294	.387	.566	.241	.372
M2MRF	IDRiD	TJDR UWF	.047	.066	.044	.096	.063
M2MRF	IDRiD	MAPLES	.220	.363	.457	.048	.272
M2MRF	IDRiD	FGADR	.010	.243	.210	.113	.144
M2MRF	IDRiD	Retinal-L.	.033	.136	.179	.068	.104
M2MRF	IDRiD	DiaRetDB1	.043	.247	.265	.107	.165
M2MRF	DDR	IDRiD	.276	.549	.678	.573	.519
M2MRF	DDR	DDR	.165	.426	.558	.456	.401
M2MRF	DDR	TJDR std	.300	.628	.664	.479	.518
M2MRF	DDR	TJDR UWF	.073	.512	.164	.075	.206
M2MRF	DDR	MAPLES	.204	.455	.429	.102	.297
M2MRF	DDR	FGADR	.015	.263	.312	.163	.188
M2MRF	DDR	Retinal-L.	.044	.148	.195	.188	.144
M2MRF	DDR	DiaRetDB1	.055	.234	.178	.247	.179
HRDecoder	IDRiD	IDRiD	.475	.699	.892	.763	.707
HRDecoder	IDRiD	DDR	.110	.355	.556	.148	.292
HRDecoder	IDRiD	TJDR std	.298	.578	.610	.426	.478
HRDecoder	IDRiD	TJDR UWF	.075	.413	.161	.359	.252
HRDecoder	IDRiD	MAPLES	.287	.444	.692	.083	.377
HRDecoder	IDRiD	FGADR	.016	.285	.338	.102	.186
HRDecoder	IDRiD	Retinal-L.	.038	.157	.338	.125	.165
HRDecoder	IDRiD	DiaRetDB1	.047	.329	.452	.175	.251
HRDecoder	DDR	IDRiD	.291	.639	.799	.633	.590
HRDecoder	DDR	DDR	.172	.557	.629	.486	.461
HRDecoder	DDR	TJDR std	.298	.707	.693	.517	.554
HRDecoder	DDR	TJDR UWF	.077	.543	.148	.102	.218
HRDecoder	DDR	MAPLES	.279	.471	.617	.159	.382
HRDecoder	DDR	FGADR	.013	.302	.291	.211	.204
HRDecoder	DDR	Retinal-L.	.064	.216	.283	.232	.199
HRDecoder	DDR	DiaRetDB1	.060	.409	.323	.306	.275

Supplementary Table 17 Foundation models without training, on every test set. Zero-shot automatic masks (density and area cap chosen on the validation pool), SAM 3 text prompts with the wording chosen on the validation pool, and prompts derived from the ground truth (an upper bound that reads the test labels). Values are mAUPR where the output carries classes and the any-lesion AUPR[a] where it does not.

Model	Setting	IDRiD	DDR	TJDR std	TJDR UWF	MAPLES	FGADR	Retinal-L.	DiaRetDB1
Zero-shot (any-lesion AUPR of the class-agnostic map; text prompts per class)									
SAM 2	zero_shot:auto_any_lesion	.023	.008	.012	.007	.003	.010	.014	.010
SAM 2	zero_shot:auto_oracle_class	.029	.009	.012	.007	.003	.010	.014	.010
SAM 2	zero_shot:auto_stability	.026	.011	.008	.018	.004	.011	.014	.008
SAM	zero_shot:auto_any_lesion	.020	.008	.008	.004	.003	.012	.014	.007
SAM	zero_shot:auto_oracle_class	.022	.008	.009	.005	.003	.013	.014	.007
SAM	zero_shot:auto_stability	.031	.014	.028	.015	.004	.020	.015	.009
Oracle boxes, every component									
MedSAM	oracle:box:all	.665	.648	.840	.860	.738	.827	.864	.714
SAM 2	oracle:box:all	.848	.756	.932	.922	.826	.825	.835	.771
SAM	oracle:box:all	.860	.772	.870	.869	.824	.759	.747	.777
Oracle points, one per component									
SAM 2	oracle:point:all	.116	.019	.081	.291	.009	.050	.058	.074
SAM 3, selected wording									
SAM 3	text:selected	.228	.125	.111	.185	.079	.108	.066	.088

[a] any-lesion AUPR of a class-agnostic map. Variant tokens as in the run records (density, cap, canvas).

Supplementary Table 18 Experiment names used in the text and the identifiers used in the repository. The run directories under runs/, the experiment column of data/results.csv and the planning documents in docs/ carry the identifiers; the text names each experiment by what it does.

Identifier	Name in the text	What it is	Where it is reported
L1	Paper audit	Extraction of the setup and the comparison tables of every method paper in the ledger	Fig. 2b,c; Supplementary Tables 1 and 2
L2	Reproducibility funnel	Registered fundus-specific methods by code, training code, weights and reproduced number	Fig. 2a
E1	Released checkpoints (author weights)	Released checkpoints scored with the paper's number, the authors' function and this protocol, on all test sets	Fig. 3a,b,d; Table 3a; Supplementary Table 16
E1r	Author-recipe reruns	The authors' own pipelines retrained from scratch on IDRiD (M2MRF, HRDecoder)	Fig. 3c; Table 3b
E2 (B-std)	Unified retraining; the unified protocol	Every tier-1 model with complete training code retrained on IDRiD and on DDR under the shared protocol	Fig. 4; Tables 4 and 5
E2b (B1)	Source-recipe track	Each model trained with its source recipe on the shared data; stopped before any run finished	Methods only
E2c	Learning-rate sweep	HRNet-W48 and M2MRF on DDR with the recipe learning rate $\times 1/3$ and $\times 3$, judged on validation only	Extended Data Fig. 7
E3a	Independent tests	Every main-table checkpoint applied to the six test sets of sources never used for training	Fig. 4b–e; Table 5; Extended Data Fig. 9
E3b	Training-source experiment	Five representative models trained on the standard-field TJDR images and on FGADR	Fig. 6b–d; Supplementary Table 7
E4	Annotation-granularity experiment	U-Net and M2MRF on fine, mixed and coarse training labels, alone and combined	Fig. 6e; Table 7; Supplementary Table 7
E5	Foundation models under four kinds of use	Zero-shot, oracle prompts, parameter-efficient fine-tuning, encoder + decoder; the encoder comparison	Fig. 5; Table 6; Supplementary Tables 10 and 17
E7	Input-resolution experiment	M2MRF and RETFound + UPerNet at a 512- or 1,440-pixel canvas and on native-resolution patches	Fig. 7d; Table 7
E8	Training-amount experiment	U-Net and M2MRF on 10%, 50% and 100% of DDR and on the merged fine-grained sources	Fig. 7e; Table 7
E9	Loss experiment	U-Net on DDR with four losses, architecture fixed	Extended Data Fig. 8; Table 7
E10	Computational cost	Parameters, trainable parameters and GPU hours of every run	Extended Data Fig. 10; Supplementary Table 8
ENS	Ensembles	Pixelwise means of the probability maps in the registered compositions	Table 5; Supplementary Table 11
REF	Label revision (Refined IDRiD)	IDRiD-trained checkpoints re-scored against the refined and the official labels	Supplementary Table 9
HUMAN	Human reference	Each DiaRetDB1 expert against the consensus of the other three	Table 5; Supplementary Table 13
CLIN	Results by DR grade	Every score recomputed within grades 0–1 and 2–4	Supplementary Table 12
N8	Reference model	Two models trained on the merged fine-grained sources and averaged; in training	Methods

Supplementary Data

- **Supplementary Data 1.** Split lists of every source (`bench/data/splits/` in the repository): the image identifiers of each train, validation and test split, the subset lists of the training-amount experiment, and the near-duplicate exclusions.
- **Supplementary Data 2.** The literature extraction (`literature/meta/reported_results.tsv`): one row per paper, baseline, dataset, class and metric, with the stated provenance of every number, and the per-paper audit (`paper_audit.tsv`).
- **Supplementary Data 3.** The results table behind every figure and table of the main text and of this document (`docs/paper/retiseg_paper/paper/data/results.csv`), with the recipe, environment and deviation record of every run; its experiment column uses the identifiers of Supplementary Table 18.
- **Supplementary Data 4.** Probability maps of every main-table model on every test set whose licence allows derived data to be released (IDRiD, DDR, TJDR, MAPLES-DR, DiaRetDB1), as 16-bit images, so that any metric can be recomputed.